

Gene Summary: Risks and Management

AUTOSOMAL RECESSIVE RISK IN CANCER GENES – MULTI-GENE PANEL TESTING

GENE & CONDITION	DESCRIPTION
ATM – Ataxia-Telangiectasia (AT)	AT is characterized by progressive cerebellar ataxia, telangiectasias, immune defects, and a predisposition to malignancy. Cells of individuals with AT are abnormally sensitive to ionizing radiation and resistant to inhibition of DNA synthesis by ionizing radiation.
BRCA1 – Fanconi anemia complementation group S (FANCS)	There are rare reports of compound heterozygous or biallelic BRCA1 P/LP variants causing FANCS. FANCS is characterized by developmental delay apparent from infancy, short stature, microcephaly, and coarse dysmorphic features. It is associated with defective DNA repair and increased chromosomal breakage.
BRCA2 – Fanconi anemia complementation group D1	FA is characterized by developmental abnormalities in major organ systems, early-onset bone marrow failure, and a high predisposition to cancer. Bone marrow failure with pancytopenia often presents in the first decade of life. Adults with biallelic BRCA2 (one allele hypomorphic) are reported. Biallelic PVs in BRCA2 are associated with early-onset acute leukemia and solid tumors.
BRIP1 – Fanconi anemia complementation group J (FANCI)	FA is characterized by developmental abnormalities in major organ systems, early-onset bone marrow failure, and a high predisposition to cancer. Bone marrow failure with pancytopenia often presents in the first decade of life.
MLH1, MSH2, MSH6, PMS2, EPCAM – CMMRD	CMMRD is a childhood cancer predisposition syndrome characterized by hematologic malignancies, brain/CNS tumors, colorectal tumors and multiple intestinal polyps, and other malignancies including embryonic tumors and rhabdomyosarcoma.
PALB2 – Fanconi anemia complementation group N (FANCN)	FA is characterized by developmental abnormalities in major organ systems, early-onset bone marrow failure, and an increased lifetime risk of cancer. Bone marrow failure with pancytopenia often presents in the first decade of life. Biallelic PVs in PALB2 are associated with solid tumors, such as medulloblastomas and Wilms tumors.

RAD51C – Fanconi anemia complementation group O	FA is characterized by developmental abnormalities in major organ systems, early-onset bone marrow failure, and a high predisposition to cancer. Bone marrow failure with pancytopenia often presents in the first decade of life.
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BRCA-Pathogenic/Likely Pathogenic Variant - Positive Management

Site	Site
General	Education regarding signs and symptoms of cancer(s), especially those associated with BRCA gene P/LP variants
Breast cancer (female)	• Breast awareness starting at age 18 years. • Clinical breast exam, every 6–12 months starting at age 25 years. • Breast screening, ➢ Age 25–29 years, annual breast MRI screening with and without contrast (or mammogram, only if MRI is unavailable) or individualized based on family history if a breast cancer diagnosis before age 30 is present. ➢ Age 30–75 years, annual mammogram and breast MRI screening with and without contrast. ➢ Age >75 years, management should be considered on an individual basis. ➢ For individuals with a BRCA P/LP variant who are treated for breast cancer and have not had a bilateral mastectomy, screening with annual mammogram and breast MRI should continue as described above. ➢ Discuss option of RRM ◇ Counseling should include a discussion regarding degree of protection, reconstruction options, and risks. In addition, the family history and residual breast cancer risk with age and life expectancy should be considered during counseling. – Address psychosocial and quality-of-life aspects of undergoing RRM. • Consider risk reduction agents as options for breast cancer, including discussion of risks and benefits
Breast cancer (male)	• Breast self-exam training and education starting at age 35 years. • Clinical breast exam, every 12 months, starting at age 35 years. • Consider annual mammogram, especially for those with BRCA2 P/LP variants in whom the lifetime risk of breast cancer is up to 7%, starting at age 50 or 10 years before the earliest known male breast cancer in the family (whichever comes first).
	• If desired, refer to fertility specialists for discussion of age-related fertility considerations, options for in vitro fertilization, egg and

Ovarian/ Fallopian Tube/ Peritoneal/ Uterine Cancers	Reproductive considerations in premenopausal women	embryo-cryopreservation, and consideration of preimplantation genetic testing, gestational carrier and adoption. • If eggs/embryos are cryopreserved, pregnancy may be achieved with uterus in place, with or without fallopian tubes or ovaries. • Individuals with P/LP BRCA1 variant may have earlier menopause and oocyte aging.
	Non-surgical risk reduction	• Consultation with gynecologic oncologist or gynecologist with expertise/experience in genetic susceptibility to gynecologic cancer recommended. • Consideration of combination estrogen/progestin (E/P) contraception (such as oral contraceptive pills [OCP]) for ovulation suppression. Overall, studies in pathogenic/likely pathogenic variant carriers support significant risk reduction benefits for ovarian cancer. • Levonorgestrel intrauterine device (LNG-IUD) has been shown to reduce risk for ovarian cancer in the average risk population
	Surgical risk reduction with bilateral salpingo-oophorectomy	• Based on age-related risks of ovarian/fallopian tube cancer: ➤ BRCA1: Recommend RRSO between 35 and 40 years ➤ BRCA2: Because ovarian cancer onset in patients with BRCA2 P/LP variants is an average of 8–10 years later than in patients with BRCA1 P/LP variants,⁸ it is reasonable to delay RRSO for management of ovarian cancer risk until age 40–45 years in patients with BRCA2 P/LP variants unless age at diagnosis in the family warrants earlier age for consideration of prophylactic surgery. • CA-125 and pelvic ultrasound are recommended for preoperative planning. • See Risk-Reducing Salpingo-Oophorectomy (RRSO) Protocol in NCCN Guidelines for Ovarian Cancer - Principles of Surgery. Appropriate surgical and pathologic expertise is strongly recommended. SEE-FIM (Sectioning and Extensively Examining the Fimbria) protocol for pathologic assessment and pelvic washings should be performed. • In addition, in premenopausal individuals, oophorectomy likely reduces the risk of developing breast cancer but the magnitude is uncertain and may be gene specific.

		• Address bone health, cardiovascular health, psychosocial health, neurologic health, sexual health, and generalized quality-of life aspects of undergoing RRSO. Consider preoperative menopause management consultation if patient is still premenopausal at time of RRSO. • Hormone replacement therapy is generally not contraindicated and thus should be discussed with premenopausal patients who do not have a personal history of breast cancer.
	Salpingectomy	• Salpingectomy reduces the risk of ovarian cancer in the general population and is an option for premenopausal patients with hereditary cancer risk who are not yet ready for oophorectomy. • Completion oophorectomy is recommended as per gene-specific guidelines. • SEE-FIM protocol for pathologic assessment and pelvic washings should be performed at salpingectomy or completion oophorectomy. • CA-125 and pelvic ultrasound are recommended for preoperative planning. • Clinical trials of interval salpingectomy and delayed oophorectomy are ongoing. Strong consideration of surgical choice study participation if available. • Consider continuation of combination OCP or hormonal IUD for continued ovarian cancer risk reduction while ovaries remain in place. • Salpingectomy is also an option for average or uncertain risk patients if they also desire surgical sterilization.
	Considerations for hysterectomy	• Limited data suggest that there may be a slightly increased risk of serous uterine cancer among individuals with a BRCA1/2 pathogenic/likely pathogenic variant. The clinical significance of these findings is unclear. Further evaluation of the risk of serous uterine cancer in the BRCA population is ongoing. The provider and patient should discuss the risks and benefits of concurrent hysterectomy at the time of RRSO for individuals with a BRCA1/2 pathogenic/likely pathogenic variant prior to surgery. • Individuals who undergo hysterectomy at the time of RRSO are candidates for estrogen-alone HRT, which is associated with a

		decreased risk of breast cancer compared to combined estrogen and progesterone, which would be required when the uterus is left in situ. • Risk of pelvic floor dysfunction or urinary incontinence after hysterectomy is influenced by factors other than hysterectomy alone; if no preceding pelvic organ prolapse, long-term follow up studies indicate risks are <5%.
	Hormone replacement options after risk reducing surgery	• In conjunction with a gynecologist or other qualified health care professional with expertise in menopause management:> ➤ HRT recommendations should be tailored depending on each patient's personal history of breast cancer and/or breast cancer risk reduction strategies. ➤ HRT is an important consideration for premenopausal patients who do not carry a diagnosis of breast cancer or do not have other contraindications for HRT. ➤ Premature menopause due to RRSO can cause detriments to bone health, cardiovascular health, psychosocial health, neurologic health, sexual health, and generalized quality-of-life. HRT can reduce these risks. • If uterus is left in place at time of RRSO, consider options for hormone replacement ➤ LNG-IUD for uterine protection with oral or transdermal estrogen. LNG- IUD may have benefits over combined HRT including potential decreased risk for breast cancer. ➤ Combination E/P HRT with counseling regarding bleeding precautions and endometrial cancer risk/awareness. ➤ Combination estrogen with selective estrogen receptor modulator (such as bazedoxifene) ➤ Combination OCPs which can be taken continuously without placebo week

Abbreviations:

ACC: adrenocortical carcinoma

AFAB: assigned female at birth

AFAP: attenuated familial adenomatous polyposis

AMAB: assigned male at birth AT ataxia-telangiectasia

BHDS: Birt-Hogg-Dube syndrome

BRRS: Bannayan-Riley-Ruvalcaba syndrome

BS: Bloom syndrome

BSE: breast self-examination

CBC: complete blood count

CHIP: clonal hematopoiesis of indeterminate potential

CLIA: Clinical Laboratory Improvement Amendments

CMMRD: constitutional mismatch repair deficiency

CNS: central nervous system

CNV: copy number variant

CPS + EG: clinical-pathologic stage + estrogen receptor status and histologic grade

CRC: colorectal cancer

CS: Cowden syndrome

ctDNA: circulating tumor DNA

DTC: direct to consumer

EUS: endoscopic ultrasound

FA: Fanconi anemia

FAP: familial adenomatous polyposis

GI: gastrointestinal

GIST: gastrointestinal stromal tumor

HDGC: hereditary diffuse gastric cancer

HLRCC: hereditary leiomyomatosis and renal cell carcinoma

HPRC: hereditary papillary renal carcinoma

HRT: hormone replacement therapy

LS: Lynch syndrome

LFS: Li-Fraumeni syndrome

MEN: multiple endocrine neoplasia

MRCP: magnetic resonance cholangiopancreatography

NF: neurofibromatosis

NGS: next-generation sequencing

P/LP: pathogenic/likely pathogenic

PCC: pheochromocytoma

PGL: paraganglioma

PHTS: PTEN hamartoma tumor syndrome

PJS: Peutz-Jeghers syndrome

PRS: polygenic risk scores

PSA: prostate-specific antigen

PV: pathogenic variant

RRM: risk-reducing mastectomy

RRSO: risk-reducing salpingo-oophorectomy

SEER: Surveillance, Epidemiology, and End Results

SNP: single nucleotide polymorphism

SNV: single nucleotide variant

UUAB: unassigned at birth

UV: ultraviolet

TPDS: tumor predisposition syndrome

TSC: tuberous sclerosis complex

VAF: variant allele frequency

VHL: Von Hippel-Lindau

VUS: variant of uncertain significance

Glossary of Relevant Genetic Terms (from the National Cancer Institute [NCI])

Autosomal dominant: Autosomal dominant inheritance refers to genetic conditions that occur when a P/LP variant is present in one copy of a given gene (i.e., the person is heterozygous).

Autosomal recessive: Autosomal recessive inheritance refers to genetic conditions that occur only when P/LP variants are present in both copies of a given gene (i.e., the person is homozygous for a P/LP variant, or carries two different variants of the same gene, a state referred to as compound heterozygosity).

de novo mutation: An alteration in a gene that is present for the first time in one family member as a result of a P/LP variant in a germ cell (egg or sperm) of one of the parents, or a P/LP variant that arises in the fertilized egg itself during early embryogenesis. Also called new P/LP variant.

Familial: A phenotype or trait that occurs with greater frequency in a given family than in the general population; familial traits may have a genetic and/or nongenetic etiology.

Family history: The genetic relationships within a family combined with the medical history of individual family members. When represented in diagram form using standardized symbols and terminology, it is usually referred to as a pedigree or family tree.

Founder effect: A P/LP variant observed with high frequency in a population founded by a small ancestral group that was once geographically or culturally isolated, in which one or more of the founders was a carrier of the mutant gene.

Germline: The cells from which eggs or sperm (i.e., gametes) are derived.

Kindred: An extended family.

Pedigree: A graphic illustration of family history.

Penetrance: A characteristic of a genotype; it refers to the likelihood that a clinical condition will occur when a particular genotype is present.

Proband: The individual through whom a family with a genetic disorder is ascertained. In males this is called a propositus, and in females it is called a proposita.

Sporadic cancer: This term has two meanings. It is sometimes used to differentiate cancers occurring in people who do not have a germline P/LP variant that confers increased susceptibility to cancer from cancers occurring in people who are known to carry a variant. Cancer developing in people who do not carry a high-risk P/LP variant is referred to as sporadic cancer. The distinction is not absolute, because genetic background may influence the likelihood of cancer even in the absence of a specific predisposing variant. Alternatively, sporadic is also sometimes used to describe cancer occurring in individuals without a family history of cancer.

Genetic/Familial High-Risk Assessment:

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Overview:

All cancers develop as a result of pathogenic or likely pathogenic (P/LP) variants in certain genes, such as those involved in the regulation of cell growth and/or DNA repair, although not all of these P/LP variants are inherited from a parent. For example, sporadic P/LP variants can occur in somatic/tumor cells only, and de novo P/LP variants can occur for the first time in a germ cell (i.e., egg or sperm) or in the fertilized egg itself during early embryogenesis. However, family studies have long documented an increased risk for several forms of cancer among first-degree relatives (i.e., parents, siblings, children) and second-degree relatives (i.e., grandparents, aunts or uncles, grandchildren, nieces, or nephews) of affected individuals. These individuals may have an increased susceptibility to cancer as the result of one or more P/LP variants present in parental germline cells; cancers developing in these individuals may be classified as hereditary or familial cancers.

Hereditary cancers are often characterized by P/LP variants associated with increased risk for certain cancers and transmission to offspring through the mother and/or father. They often have an early age of onset and exhibit an autosomal dominant inheritance pattern (i.e., occur when the individual has a P/LP variant in only one copy of a gene). Familial cancers share some but not all features of hereditary cancers. For example, although familial breast cancers occur in a given family more frequently than in the general population, they generally do not exhibit the inheritance patterns or onset age consistent with hereditary cancers. Familial cancers may be associated with chance clustering of sporadic cancer cases within families, genetic variation in lower penetrance genes, a shared environment, or combinations of these factors.

An individual suspected of being at risk for hereditary cancer should be offered genetic counseling. This is consistent with recommendations from the U.S. Preventive Services Task Force (USPSTF). Assessment of an individual’s risk for familial or hereditary cancer is based on a thorough evaluation of the personal and family history. With respect to hereditary cancers, advances in molecular genetics have identified a number of genes associated with inherited susceptibility to breast, ovarian, and pancreatic cancer (e.g., BRCA1/2, PALB2, ATM) and have provided a means of characterizing the specific P/LP variant present in certain individuals and families exhibiting an increased risk for cancer. The field of cancer genetics has implications for all aspects of cancer-related care of individuals with hereditary or familial cancers, including prevention, screening, and treatment.

The NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, and Pancreatic were developed with an acute awareness of the preliminary nature of much of our knowledge regarding the clinical application of the rapidly emerging field of molecular genetics, and with an appreciation for the need for flexibility when applying these guidelines to individual families. Furthermore, it should be emphasized that these Guidelines were not developed as a substitute for professional genetic counseling. Rather, they are intended to: 1) serve as a resource for health care providers to identify individuals who may benefit from cancer risk assessment and genetic counseling and testing; to guide decisions related to genetic testing; and 3) facilitate a multidisciplinary approach in the comprehensive care of individuals at increased risk for hereditary breast, ovarian, and pancreatic cancer. The current NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate focus primarily on assessment of P/LP variants associated with increased risk of breast, ovarian, pancreatic, and prostate cancer, including BRCA1, BRCA2, CDH1, PALB2, PTEN, and TP53, and recommended approaches to genetic counseling/testing and care strategies in individuals with these P/LP variants. Where possible, P/LP variants in more recently identified genes have been addressed to the extent possible given the limited information available. Recommendations regarding P/LP variants associated with pancreatic cancer, and pancreas screening for individuals harboring such variants, were added to the NCCN Guidelines in the 2020 update. Additionally, testing criteria for those with or at risk for prostate cancer have also been included in the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®), Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate, 2025.)

Genetic Risk Assessment and Counseling:

Cancer genetic risk assessment and genetic counseling is a multi-step process involving the identification and counseling of individuals at risk for familial or hereditary cancer. The purpose of cancer genetic counseling is to educate individuals about the genetic, biological, and environmental factors related to a cancer diagnosis and/or risk for disease to help derive personal meaning from cancer genetic information, and to empower them to make educated, informed decisions about genetic testing, cancer screening, and cancer prevention. Many patients undergoing genetic testing do not receive proper counseling. Further, testing rates are inadequate among some populations with higher risk, such as African American individuals. A genetic counselor, clinical geneticist, oncologist, surgeon, oncology nurse, or other health professional with expertise and experience in cancer genetics should be involved in every stage of the process.

Testing is clinically indicated in individuals for whom there is a personal or family history suggesting genetic cancer susceptibility and for whom results will aid in risk management and treatment. The selection of genes for which testing is indicated is based on the personal and familial characteristics that determine the individual's prior probability of being a carrier of a P/LP variant, and on the psychosocial

degree of readiness of the person to receive genetic test results. Genetic risk assessment is a dynamic process and can change if additional relatives are diagnosed with cancer. The genetic testing strategy is greatly facilitated when a P/LP variant has already been identified in another family member. In that case, the genetic testing laboratory can limit the search for P/LP variants in additional family members to the same location in the gene. However, if there is reason to suspect more than one P/LP variant in the family, then broader testing may be considered.

For the majority of families in whom presence of a P/LP variant is unknown, it is best to consider testing an affected family member first, especially a family member with early-onset disease, bilateral disease, or multiple primaries, because that individual has the highest likelihood of a positive test result. The testing of the unaffected individual (or of unaffected family members) is reasonable when no affected family member is available for testing. In such cases, it is most informative to test the unaffected individual or unaffected close relative with the highest likelihood of testing positive for the P/LP variant. This may include the relative closest to the family member with the youngest age at diagnosis, bilateral disease, multiple primary tumors, or other cancers associated with a suspected hereditary syndrome. A negative test result in such cases, however, is considered indeterminate and does not provide the same level of information as when there is a known P/LP variant in the family. Thus, one should be mindful that, when testing unaffected individuals (in the absence of having tested affected family members), significant limitations may exist in interpreting the test results, and testing multiple family members may be indicated since absence of a P/LP variant in one unaffected relative does not rule out a P/LP variant in other family members. The maternal and paternal sides of the family should be considered independently for familial patterns of cancer. “Limited” family structure is defined as two or fewer first- or second-degree female relatives who survive past age 45 (on either side of the family) and/or possessing no or inadequate information about one’s birth parents.

Individuals who have received allogeneic hematopoietic cell transplantation (HCT) should not have molecular genetic testing performed on blood samples. In such cases, DNA of the individual being tested should be extracted from a fibroblast culture, if available. If this is not possible, buccal cells or saliva may be considered as an alternative source for DNA; however, a study has reported that over time, buccal epithelial cells are replaced by donor-derived cells in allogeneic HCT recipients. Therefore, genetic testing using saliva or buccal swab samples may be limited given this known risk of contamination or malignant cells from the hematologic malignancy. Fibroblasts are also indicated when testing individuals with active or recent hematologic malignancies.

A counseling dilemma is posed by the finding of a variant of uncertain significance (VUS), a genetic alteration that may actually represent a benign polymorphism unrelated to an increased cancer risk or may indicate an increased cancer risk. Retrospective analyses have estimated that 82.1% to 91.2% of the time, if a VUS is reclassified, it is downgraded to benign or likely benign, while 8.7% to 17.9% are

upgraded from VUS to pathogenic. Therefore, VUS should not be used to alter medical management. These patients should be considered for referral to research studies that aim to define the functional impact of the gene variant, such as variant reclassification programs through clinical labs or registries. Some examples of these programs and registries include ClinVar (the archival database at the National Center for Biotechnology Information [NCBI]); the NIH-funded Clinical Genome Resource (ClinGen; <https://www.clinicalgenome.org/>); the international Evidence-based Network for the Interpretation of Germline Mutant Alleles (ENIGMA; <https://enigmaconsortium.org/>); and the International Society for Gastrointestinal Hereditary Tumors (InSIGHT; <http://insight-group.org/>). It is important to note that there may be inconsistencies among how some programs and registries interpret the clinical actionability of some VUS, and there are discordant variant interpretations across laboratories. These inconsistencies may lead to confusion regarding medical management, and careful counseling and skilled interpretation are required. RNA studies (when appropriate) may be a consideration to further define functional impact of variants. Family members should not be tested for a VUS for the purposes of clinical management unless there are conflicting data between laboratories regarding the classification of a variant. In the event where there are discrepancies in classification, careful consideration must be taken regarding family history, testing family members, and if other functional studies could aid in variant classification. Clinicians and scientists should work together to develop a VUS classification system as more information is discovered in research studies. Risk management strategies in carriers of a VUS or likely benign variant should be based on family history of cancer.

Carriers of a P/LP variant should be encouraged to participate in clinical trials or genetic registries. Carriers should be encouraged to recontact their genetics providers every few years for updates, as laboratories may issue amended reports as the knowledge base surrounding hereditary cancer risk expands.

Tumor Genomic Testing:

Tumor profiling can be considered complementary to germline testing. However, the absence of a P/LP variant for a given gene from tumor profiling does not rule out the possibility of a germline P/LP variant in that gene. Tumor genomic testing tends to be designed to address treatment actionability and prognosis. Therefore, a variant interpreted as P/LP in the germline may be interpreted as normal or as a VUS in the tumor, if that variant has no clear clinical implications. In addition, the sensitivity of most tumor testing is lower (particularly for intermediate-sized deletions and duplications) than that for most dedicated germline tests, sometimes due to filtering out of germline findings reported in tumor sequencing results. In a study of 21,333 patients with cancer who underwent both tumor and germline testing at an National Comprehensive Cancer Network® (NCCN®) Member Institution, tumor-only sequencing missed 10.5% of clinically actionable P/LP variants. If a patient meets testing criteria for

germline testing for a given gene, then confirmatory germline testing should be considered through a CLIA-approved lab despite tumor profiling results. Circulating tumor DNA (ctDNA) assays may be used by some labs. ctDNA has the potential to identify both somatic and germline variants. However, since the primary intent of tumor testing is to inform treatment decision-making, ctDNA assays are not validated for reporting or interpretation of germline variants. The sensitivity, false-positive rates, and positive predictive value of ctDNA tests for early-stage disease, which are needed to derive clinical utility and to determine clinical validity, are also not fully defined. The psychological impact of ctDNA testing also remains unknown. If a germline variant that could impact medical management is detected with a ctDNA assay, then confirmatory testing with a CLIA-approved assay intended for detection and interpretation of germline results is recommended.

Multi-Gene Testing:

Next-generation sequencing allows for the sequencing of multiple genes simultaneously. This is referred to as multi-gene testing. Multi-gene testing can detect P/LP variants not found in single-gene testing.³⁶⁻³⁸ Since more than one gene can explain an inherited cancer syndrome, phenotype directed testing based on personal and family history through a multi-gene panel test is often more efficient and/or cost-effective. Multi-gene testing may also be considered for those who tested negative for one particular syndrome, but whose personal and family history is suggestive of an inherited susceptibility. It is now common practice to order multigene panel tests that include genes beyond the original indication for which testing is warranted. Phenotype needs to be considered when ordering multi-gene panel tests, to ensure that the relevant genes are included.

There are several issues to consider regarding multi-gene testing. First, commercially available tests may differ significantly on a number of factors, such as number of genes analyzed, turnaround time, insurance coverage, laboratory expertise, variant reclassification protocol, methods of DNA/RNA analysis, and availability of financial assistance for cascade testing of relatives, among others. Therefore, the specific laboratory and multi-gene test should be chosen carefully. In addition, P/LP variants identified for more than one gene add complexity that may lead to difficulty in making risk management recommendations. A management plan based on genetic test results should only be developed for identified P/LP variants that are clinically actionable.

A major dilemma regarding multi-gene testing is that there are limited data and a lack of clear guidelines regarding degree of cancer risk associated with some of the genes assessed, and how to communicate and manage risk for carriers of these genes.⁴⁴⁻⁴⁸ This issue is compounded by the low incidence rates of hereditary disease, leading to a difficulty in conducting adequately powered studies.⁴⁴ Multi-gene tests include moderate penetrance genes, and they often also include low-penetrance genes for which there are little available data regarding degree of cancer risk and guidelines

for risk management.^{39,49} Analysis from a prospective, multicenter cohort study including 2984 patients with cancer unselected based on cancer type, disease stage, family history of cancer, age of diagnosis, and ethnicity showed that, with use of an 80-gene panel test, a P/LP variant was found in 13.3%, with a highly penetrant variant found in 5%.⁵⁰ About half of the identified variants were of moderate or low penetrance, and a VUS was also found in about half the sample. In another study utilizing a 21-gene panel in 9714 patients diagnosed with multiple primary cancers, a P/LP variant was found in 13.6%, with P/LP variant frequency increasing with number of primaries diagnosed ($P = .00056$).⁵¹ The use of tailored panels that are disease-focused and include clinically actionable cancer susceptibility genes is preferred over large panels that include genes of uncertain clinical relevance. Also, certain variants in a gene may be associated with a different degree of risk than other variants in that gene. For example, the presence of the c.7271T>G missense P/LP variant in ATM is associated with an increased risk for early-onset breast cancer.

Multi-gene tests also increase the likelihood of detecting a VUS.^{37-39,45,55-57} An analysis of germline genetic testing results through 2019 for 200,000 patients with breast cancer and 15,000 patients with ovarian cancer diagnosed between 2013 and 2017 showed that VUS rates increased from 2013 to 2017 for both the patients diagnosed with breast cancer (8.5%–22.4%) and the patients diagnosed with ovarian cancer (8.1%–28.3%).⁵⁸ This study also demonstrated racial and ethnic differences in VUS rates, with Asian, Black, and Hispanic patients having significantly higher VUS rates than white patients in 2017 ($P < .001$), for both breast and ovarian cancer. There are mixed data about the potential for harm due to misinterpretation of VUS results (e.g., over-treatment), with a 2017 study reporting that half of patients with VUS who were not considered increased risk for carrying a P/LP variant undergo risk-reducing mastectomy (RRM), suggesting potential overtreatment. In contrast, a 2021 meta-analysis including 22 studies showed that individuals carrying a P/LP variant had higher rates of risk-reducing surgery compared to individuals with a VUS, indicating that individuals with a VUS may not be very commonly undergoing inappropriate risk management. There is also an increase in the chance of finding genotypically distinct cell lines (i.e., genetic mosaicism) with next-generation sequencing. Clones of non-cancerous cells (i.e., aberrant clonal expansion) containing a P/LP TP53 variant have been found in healthy adults undergoing multi-gene testing. This phenomenon can often be attributed to clonal hematopoiesis, a condition in which a hematopoietic stem cell begins making blood cells with the same acquired P/LP variant. When there is no evidence of a hematologic malignancy, then it is referred to as clonal hematopoiesis of indeterminate potential (CHIP). Age-related CHIP is associated with increased risk of hematologic malignancies but may also lead to unnecessary clinical intervention. Ancillary testing of non-lymphoid noncancerous tissue can be used to help determine the true presence of a germline variant.

Polygenic risk scores (PRS) are now sometimes included in some genetic test reports. PRS are groups of SNPs associated with a specific disorder or disease, such as cancer. Studies evaluating the validity

of PRS to refine risks in those with hereditary cancer have been conducted primarily with breast and prostate cancers. Two studies identified PRS that were strongly associated with ER-negative breast cancer in carriers of a BRCA1 P/LP variant, overall breast cancer in carriers of a BRCA2 P/LP variant, and high-grade serous ovarian cancer in carriers of both BRCA1 and BRCA2 P/LP variants. Two studies of male carriers of a BRCA1/2 P/LP variant identified PRS associated with breast cancer risk and prostate cancer risk. Studies have also evaluated the potential clinical utility of incorporating PRS into a risk-stratified approach for screening for prostate cancer and for identifying age of onset of aggressive prostate cancer. Studies of PRS have largely been done with those of European ancestry. Studies with larger samples from diverse populations are needed. Given that the clinical value of PRS has not yet been established, these should not be used to inform clinical management at this time.

Pre- and Post-Test Counseling:

For individuals potentially meeting established criteria for one or more of the hereditary cancer syndromes, genetic testing should be considered along with appropriate pre- and post-test counseling. Pre-test counseling should include a discussion of why the test is being offered and how test results may impact medical management, cancer risks associated with the P/LP variant in question, the significance of possible test results (positive, true negative, uninformative negative, VUS, mosaic results; see Principles of Cancer Risk Assessment and Counseling in the algorithm for the complete definitions of these terms), the likelihood of a positive result, technical aspects and accuracy of the test, cost considerations, risks of genetic discrimination, psychosocial aspects, confidentiality issues, implications for treatment decision-making, the potential significance of the test results for family members, and other topics. A discussion of confidentiality issues should include an explanation of the federal Genetic Information Non-discrimination Act (GINA) enacted in 2008, which prohibits most health insurers and employers from discrimination on the basis of genetic test results. Since some patients with cancer who have a poor prognosis may be unable to receive results directly, a plan for results disclosure should be discussed, such as the patient consenting to Release of Information of test results to a spouse or other close relative. A detailed family history should be collected, which involves development of an expanded pedigree, beginning with the health of the individual diagnosed with cancer and proceeding outward to include first-, second-, and third-degree relatives on both the maternal and paternal sides. Factors that limit the informativeness of the pedigree are small family size, a small number of individuals of the susceptible gender for sex-limited cancers, reduced penetrance, early deaths in family members (which precludes the possibility that they will develop adult diseases), prophylactic surgeries that remove an organ from subsequent risk for cancer (e.g., hysterectomy for uterine fibroids in which the ovaries are also removed), adoptions, and inaccurate or incomplete information on family members (e.g., in the case of adoption or divorce). It is also important to know

the ancestry/ethnicity of the individual, since members of certain groups (e.g., Ashkenazi Jewish) have increased risks of carrying P/LP variants for specific diseases. Any family members who received genetic testing should also be noted, as well as testing results. Finally, a detailed medical and surgical history from the proband should be collected, and a physical examination should be performed by a qualified clinician when appropriate.

The presentation of testing information is most effective when tailored to the age and education of the person undergoing counseling, and that individual's personal exposure to the disease, level of risk, and social environment. Information could be delivered in person or over the phone. Telehealth (i.e., real-time two-way videoconference) is also increasingly utilized as a feasible alternative for in-person genetic counseling. Remote options (e.g., telephone, telehealth) have the potential to help improve genetic testing rates in areas with inadequate access.

Post-test counseling includes disclosure of results, a discussion of the associated medical risks, an assessment of the impact of the results on the emotional state of the individual, a discussion of the impact of the results on the comprehensive care of the individual (including discussion of therapeutic implications by a qualified health care professional), and how and where the patient will be screened for cancer risk. Counseling should include making the individual aware of any available resources, such as disease-specific support groups, high-risk clinics, advocacy groups, and research studies. The counselor should discuss the importance of genetic counseling and testing for relatives who also may be at increased risk.

Since some P/LP variants are associated with rare autosomal recessive conditions (e.g., Fanconi anemia is associated with BRCA2, BRIP1, and PALB2 variants), the proband should be advised regarding possible inherited cancer risk to relatives and their own options for risk assessment and management. Testing of a partner of a carrier of a P/LP variant may also be considered to inform reproductive decision-making. See Autosomal Recessive Risk in Cancer Genes – Multi-Gene Panel Testing in the algorithm for a full list of the P/LP variants covered in these Guidelines that are associated with autosomal recessive conditions.

Pre- and post-test genetic counseling with involvement of an expert in cancer genetics is recommended. However, the panel acknowledges that most genetic testing is conducted by providers with limited expertise in genetics and often without pre-test genetic counseling. Shortages in genetics health providers, expansion of testing indications, aggressive marketing, and increased accessibility of testing due to plummeting costs inclusive of DTC models for testing provide the impetus for the panel to identify scenarios in which referral to a genetics health provider should be considered. These scenarios are as follows: identification of a P/LP variant; negative results despite tumor profiling, personal history, or family history suggestive of inherited condition; VUS result that warrants further evaluation or for which a patient or provider considers using to guide management; mosaic/possibly mosaic result or

clonal hematopoiesis (CH); discrepant interpretation of variants (e.g., discordant results across laboratories); interpretation of PRS; and detection of P/LP variants from DTC testing.

Many patients who have been diagnosed with cancer and have a P/LP variant are at increased risk for additional primary cancers in the future. Management of those risks may be appropriate after treatment of the current cancer or may be combined with treatment for a current cancer. For example, a patient with breast cancer and a pathogenic variant in BRCA1 or BRCA2 may consider bilateral mastectomies to treat their current cancer and also to reduce the risk of a future primary breast cancer. These patients may also consider oophorectomy for treatment of hormone receptor-positive breast cancer and also for reducing ovarian cancer risk.

Best practices for communicating an individual's personal risk relative to published estimated lifetime risks of cancer include the following: 1) presenting risk estimates as a range rather than a single estimate (e.g., 30% to 40%); 2) presenting absolute risk versus relative risk terminology; 3) acknowledging the margin of error associated with risk estimates and how these are impacted by number of individuals with a P/LP variant; and 4) acknowledging that risk estimates can change over time. Specifically, patients who are older will have lower remaining lifetime risks. Over time, patients with a P/LP variant benefit from re-consultation with a medical provider who is familiar with inherited risk for cancer. This re-consultation is important for the following reasons: 1) increases compliance with screening guidelines, since screening behaviours may decrease over time; 2) allows the patient to re-evaluate personal choices about risk-reducing surgeries, based on changing life stage and circumstances; and 3) provides opportunities to “check in” with the patient about following up-to-date risk management guidelines, discuss additional or emerging genetic testing options, and review improved risk models. Cancer risk estimates can change based on larger case-control studies. Similarly, recommendations for screening and risk reduction can change based on new technologies and data. The frequency of follow-up with the patient will depend on factors such as age, reproductive planning, comorbidities, risk reducing surgeries, and others as applicable.

Counseling may be warranted for those with negative or indeterminate results, as reasons for a negative result include the following scenarios: P/LP variant exists in a gene variant that was not recognized due to limitations in technology; P/LP variant exists in a gene variant that was not evaluated; and potential presence of a P/LP variant in a family member that was not detected in the individual. The determination of a “true negative” result depends on the specific family history of cancer, the specific P/LP variant found, and the relationship to any family members who test positive. When an individual has tested negative, it may still be appropriate to consider increased screening and risk reduction measures for cancer, based on family history. Over time, this individual may be a candidate for additional genetic testing due to additional family history or as new genes are identified to be associated with cancer risk or as technology advances.

Reproductive Options

The outcomes of genetic testing can have a profound impact on family planning decisions for individuals of reproductive age who are found to be carriers of a P/LP variant. Counseling for reproductive options such as prenatal diagnosis and assisted reproduction using preimplantation genetic testing (PGT) and donor gametes may therefore be warranted for couples expressing concern over their future offspring's carrier status of a P/LP variant. Such counseling should include a comprehensive discussion of reproductive options, extent of cancer risk balanced with cancer worry, degree of protection for breast, ovarian and uterine cancer, management of menopausal symptoms, hormone replacement therapy (HRT), related medical or surgical history, and consideration of a gestational carrier.

Prenatal diagnosis involves post implantation genetic analysis of an early embryo, utilizing chorionic villi or amniotic fluid cell samples; genetic testing is typically conducted between week 12 and week 16 of gestation, and testing results may potentially lead to a couple's decision to terminate the pregnancy. PGT has emerged as an alternative method of genetic testing in early embryos. PGT involves the testing of 1 or 2 cells from embryos in very early stages of development (i.e., 6–8 cells) after in vitro fertilization (IVF). This procedure allows for the selection of unaffected embryos to be transferred to the uterus and may therefore offer the advantage of avoiding potential termination of pregnancy. The PGT process requires the use of IVF regardless of the fertility status of the couple (i.e., also applies to couples without infertility issues), and IVF may not always lead to a successful pregnancy. Lastly, the technology or expertise may not be readily available in a couple's geographic location. If eggs/embryos are cryopreserved, pregnancy may be achieved with uterus in place, with or without fallopian tubes or ovaries.

Various factors, both medical and personal, must be weighed in the decision to utilize prenatal diagnosis or PGT. Medical considerations may include factors such as the age of onset of the hereditary cancer, penetrance, severity or associated morbidity and mortality of the cancer, and availability of effective cancer risk reduction methods or effective treatments. For example, results from two systematic reviews have suggested that carrying a P/LP BRCA1/2 variant may be associated with diminished ovarian reserve. Although the use of prenatal diagnosis or PGT is relatively well established for severe hereditary disorders with very high penetrance and/or early onset (e.g., Fanconi anemia), its use in conditions associated with lower penetrance and/or later onset (e.g., hereditary breast or ovarian cancer syndrome) remains somewhat controversial from both an ethical and regulatory standpoint. Personal considerations for the decision to utilize prenatal diagnosis or PGT may include individual ethical beliefs, value systems, cultural and religious beliefs, and social and economic factors. Successful births have been reported with the use of PGT and IVF in carriers of a BRCA1/2 P/LP variant, but data in the

published literature are still very limited. In addition, data pertaining to long-term safety or outcomes of PGT and assisted reproduction in carriers of a P/LP variant are not yet available.

High-Penetrance Breast and/or Ovarian Cancer Susceptibility Genes:

Specific patterns of hereditary breast and ovarian cancers have been found to be linked to P/LP variants in the BRCA1/2 genes. In addition, two very rare hereditary cancer syndromes exhibiting an increased risk for breast cancer are Li-Fraumeni syndrome (LFS) and Cowden syndrome, which are related to germline P/LP variants in the TP53 and PTEN genes, respectively. PALB2, STK11, and CDH1 are also considered high penetrance breast cancer susceptibility genes. These hereditary syndromes share several features beyond elevation of breast cancer risk. These syndromes arise from germline P/LP variants that are not within sex-linked genes; hence, the variants can be inherited from either parent. The syndromes are associated with breast cancer onset at an early age and development of other types of cancer and exhibit an autosomal dominant inheritance pattern (see Table 1). Offspring of an individual with one of these hereditary syndromes have a 50% chance of inheriting the P/LP variant. In addition, individuals with these hereditary syndromes share increased risks for multiple cases of early-onset disease as well as bilateral disease. The P/LP variants associated with these hereditary syndromes are considered to be highly penetrant. In addition, the manifestations (i.e., expression) of these hereditary syndromes are often variable in individuals within a single family (e.g., age of onset, tumor site, number of primary tumors). The risk of developing cancer in individuals with one of these hereditary syndromes depends on numerous variables including the gender and age of the individual.

Prior to 2020, the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast and Ovarian (Breast, Ovarian, and Pancreatic as of 2020) focused largely on testing criteria for BRCA1/2 and appropriate risk management for carriers of a BRCA1 or BRCA2 P/LP variant. Sections on LFS and Cowden syndrome/PTEN hamartoma tumor syndrome (PHTS) were also included. Based on strong evidence that genes beyond BRCA1/2, TP53, and PTEN confer markedly increased risk of breast and/or ovarian cancers, these Guidelines have been expanded; see the sections below on other P/LP variants associated with breast/ovarian cancer.

BRCA-Related Breast/Ovarian Cancer Syndrome:

Both the BRCA1 and BRCA2 genes encode for proteins involved in tumor suppression. BRCA1/2 P/LP variants can be highly penetrant (for definition, see Table 1), although the probability of cancer development in carriers of BRCA1/2 P/LP variants is variable, even within families with the same variant. At present, it is unclear whether penetrance is related only to the specific P/LP variant identified in a family or whether additional factors, either genetic or environmental, affect disease expression.

Epigenetic modification can also influence disease penetrance for a P/LP variant. It is generally accepted, however, that carriers of BRCA1/2 P/LP variants have an excessive risk for both breast and ovarian cancer that warrants consideration of more intensive screening and preventive strategies.

Breast Cancer Risk:

Estimates of penetrance for lifetime risk for primary breast cancer range from 60% to 72% for carriers of a *BRCA1* P/LP variant and 55% to 69% for

carriers of a *BRCA2* P/LP variant. Risk for contralateral breast cancer in carriers of a *BRCA2* P/LP variant is age-dependent and greatest in those diagnosed with breast cancer at an early age (i.e., <40 years). Risk is also greater in *BRCA1* P/LP variant carriers compared to *BRCA2* P/LP variant carriers (20-year cumulative risk 30%–40% and 25%, respectively). A study of UK Biobank data showed that, among carriers of a *BRCA1/2* P/LP variant, breast cancer risk was greater in those with a first-degree relative diagnosed with breast cancer, compared to those who did not have this family history.

While the evidence is mixed, we do not currently have evidence to support that BRCA-associated breast cancers are more aggressive and/or have poorer outcomes. A meta-analysis including 13 studies showed that carriers of a BRCA1 P/LP variant with breast cancer had worse overall survival (OS) compared to those without a BRCA1 or BRCA2 P/LP variant (hazard ratio [HR], 1.50; 95% CI, 1.11–2.04), while OS did not significantly differ between those harboring a BRCA2 P/LP variant and those without a BRCA1 or BRCA2 P/LP variant (HR, 0.97; 95% CI, 0.78–1.22). Another meta-analysis including 60 studies and 105,220 patients with breast cancer also found that carriers of a BRCA1 P/LP variant had worse OS compared to non-carriers (HR, 1.30; 95% CI, 1.11–1.52; $P = .001$). Carriers of a BRCA2 P/LP variant had worse breast cancer-specific survival compared to non-carriers (HR, 1.29; 95% CI, 1.03–1.62; $P = .03$), though OS was not significantly different. This meta-analysis also showed that, among patients with triple-negative breast cancer, BRCA1/2 P/LP variants are associated with better OS (HR, 0.49; 95% CI, 0.26–0.92; $P = .03$). However, this subgroup analysis only included two studies. A third meta-analysis including 66 studies also showed that a BRCA2 P/LP variant was associated with worse breast cancer-specific survival (HR, 1.57; 95% CI, 1.29–1.86), but study results were too heterogeneous for the analysis to be conclusive.¹⁰⁹ Results of the prospective cohort Prospective Outcomes in Sporadic versus Hereditary breast cancer (POSH) study including 2733 females with breast cancer showed no significant differences in OS between carriers of a BRCA1/2 P/LP variant and non-carriers 2, 5, and 10 years after diagnosis.

BRCA1/2 P/LP variants are associated with early-onset primary breast cancer. In a sample of 21,401 families who met German Consortium for Hereditary Breast and Ovarian Cancer testing criteria for *BRCA1/2* P/LP variants, a P/LP variant was detected in 13.7% of families with a single case of breast cancer diagnosed at <36 years of age. An analysis of 6478 patients who were diagnosed with breast

cancer before 50 years of age showed that carriers of a *BRCA1* P/LP variant had worse OS compared to patients who were not carriers of a P/LP *BRCA1/2* variant (HR, 1.28; 95% CI, 1.05–1.57; $P = .01$), but this association was no longer statistically significant when taking into account disease and treatment characteristics (HR, 1.20; 95% CI, 0.97–1.47; $P = .09$). *BRCA2* P/LP variants were not significantly associated with decreased OS in these analyses, except for the first 5 years of follow-up (HR, 1.56; 95% CI, 1.06–2.28; $P = .02$).

Some histopathologic features have been reported to occur more frequently in breast cancers of individuals with a germline *BRCA1/2* P/LP variant. For example, several studies have shown that *BRCA1*-related breast cancer is more likely to be characterized as ER-/PR-negative and HER2-negative (i.e., “triple negative”). Studies have reported *BRCA1* P/LP variants in 4.4% to 16% of patients with triple-negative breast cancer. One cohort study showed an absolute lifetime risk of 40% for hormone receptor-positive (ER+ and/or PR+) breast cancer in carriers of a P/LP *BRCA2* variant. The case-control BRIDGES study showed an association between P/LP *BRCA2* variant and increased risk for hormone receptor-positive (ER+ and/or PR+) HER2-negative breast cancer (odds ratio [OR], 11.53; 95% CI, 8.92–14.90), though a significant association was also identified for triple negative breast cancer (OR, 10.07; 95% CI, 7.61–13.32). The Breast Cancer Association Consortium and the CARRIERS case-control studies showed associations between a *BRCA2* P/LP variant and increased risk of ER-positive breast cancer (1.46%; OR, 5.68; 95% CI, 4.65–6.96 and 1.09%; OR, 4.66; 95% CI, 3.52–6.23, respectively). Another case-control study showed that the 20-year survival rate in carriers of a *BRCA2* P/LP variant with ER-positive tumors was 62.2%, compared to 83.7% in those with ER-negative tumors, though this difference was only statistically significant in those association between *BRCA2* P/LP variant with ER-positive tumors and poor survival outcomes is currently unknown and warrants investigation, though one hypothesized explanation includes difference in estrogen signaling pathways and increased sensitivity to ovarian hormones for these tumors.

Among patients with triple-negative disease, carriers of a P/LP *BRCA* variant were diagnosed at a younger age compared with non- carriers. In a study of a large cohort of patients with triple-negative breast cancer ($N = 403$), the median age of diagnosis among carriers of a P/LP *BRCA1* variant ($n = 65$) was 39 years. Patients in this population- based study were unselected for family history or age. Among the group of patients with early-onset (age at diagnosis <40 years) triple-negative breast cancer ($n = 106$), the incidence of *BRCA1* P/LP variants was 36%; the incidence was 27% among those diagnosed before 50 years of age ($n = 208$). Results from the prospective cohort POSH study showed that, among 558 patients with triple-negative breast cancer, 2-year OS was greater in carriers of a *BRCA1/2* P/LP variant than in non-carriers (95% vs. 91%, respectively; HR, 0.59; 95% CI, 0.35–0.99; $P = .047$), but 5- and 10- year OS did not differ significantly between these groups. A SEER analysis from California and Georgia including females diagnosed with breast cancer showed that, among 5461 diagnosed with

triple-negative breast cancer, cancer-specific mortality was lower among carriers of a P/LP *BRCA1* variant (HR, 0.49; 95% CI, 0.35–0.69) and carriers of a P/LP *BRCA2* variant (HR, 0.60; 95% CI, 0.41–0.89), compared to non- carriers.

Carriers of a P/LP *BRCA1/2* variant who were assigned male at birth also have a greater risk for cancer susceptibility. Among male patients with breast cancer unselected for family history, 4% to 14% tested positive for a germline *BRCA2* P/LP variant. For males carrying a P/LP *BRCA2* variant, the cumulative lifetime risk for breast cancer has been estimated at 1.8% to 7.1%. The cumulative lifetime risk for male carriers of a P/LP *BRCA1* variant is 0.2% to 1.2%. In contrast, for males who are not carriers of a P/LP *BRCA1/2* variant, the lifetime risk for breast cancer has been estimated at approximately 0.1% (1 in 1000).

Ovarian Cancer Risk

Increased risks for cancers of the ovary, fallopian tube, and peritoneum are observed in carriers of a P/LP *BRCA1/2* variant. In the setting of an invasive ovarian cancer diagnosis, a P/LP *BRCA1* variant has been found in 3.8% to 14.5% of patients, and a P/LP *BRCA2* variant has been found in 4.2% to 5.7% of patients. *BRCA1* variants have an estimated 48.3% (95% CI, 38.8%–57.9%) cumulative risk of ovarian cancer by age 70, while the cumulative risk by age 70 is 20.0% (95% CI, 13.3%–29.0%) for carriers of a P/LP *BRCA2* variant.

Several studies have reported more favorable survival outcomes among carriers of a P/LP *BRCA1/2* variant in patients with ovarian cancer compared with patients who are non-carriers. Survival outcomes appear to be most favorable for carriers of a P/LP *BRCA2* variant. A systematic review and meta-analysis including 13 references with 4565 patients (1131 *BRCA1/2* P/LP variant carriers) showed that 5-year survival was higher in the P/LP carriers than in non-carriers (risk difference, 14.9%; $P = .0002$; relative risk [RR], 1.36; $P = .001$), but the difference in risk was less pronounced for 10-year survival (risk difference, 8.6%; $P = .042$; RR, 1.25; $P = .12$). *BRCA2* P/LP variants are also associated with significantly higher response rates (compared with non-carriers or with carriers of a *BRCA1* P/LP variant) to primary chemotherapy. In contrast, *BRCA1* P/LP variants were not associated with prognosis or improved chemotherapy response.

The histology of ovarian cancers in carriers of a P/LP *BRCA1/2* variant is more likely to be characterized as serous adenocarcinoma and high grade compared with ovarian cancers in non-carriers, although endometrioid and clear cell ovarian cancers also have been reported in the former population. P/LP variants are also associated with non- mucinous ovarian carcinoma as opposed to mucinous. Mucinous epithelial ovarian carcinomas may be associated with other P/LP variants, such

as *TP53*, which are implicated in LFS (see below). Non-epithelial ovarian carcinomas (e.g., germ cell and sex cord-stromal tumors) are not significantly associated with a *BRCA1/2* P/LP variant,¹⁶⁸ though ovarian sex cord tumor with annular tubules is associated with *STK11* P/LP variants. Current data show that ovarian low malignant potential tumors (i.e., borderline epithelial ovarian tumors) are also not associated with a *BRCA1/2* P/LP variant.

In studies of carriers of a P/LP *BRCA1/2* variant who underwent risk-reducing salpingo-oophorectomy (RRSO) for occult gynecologic neoplasia, both invasive carcinoma and intraepithelial lesions were identified in 4.5% to 9% of cases based on rigorous pathologic examinations of the ovaries and fallopian tubes. Tubal intraepithelial carcinoma (TIC) is thought to represent an early precursor lesion for serous ovarian cancers, and TIC (with or without other lesions) was detected in 5% to 8% of cases from patients carrying a P/LP *BRCA1/2* variant who underwent RRSO. The fimbriae or distal tube was reported to be the predominant site of origin for these early malignancies found in carriers of a P/LP *BRCA1/2* variant. Although TIC appeared to present more frequently among carriers of a P/LP *BRCA1/2* variant compared with non-carriers undergoing RRSO, TIC has also been documented among patients with serous carcinomas unselected for family history or *BRCA* P/LP variant status. Because TIC was identified in individuals who underwent surgery for risk reduction (for carriers of a P/LP *BRCA1/2* variant) or other gynecologic indications, the incidence and significance of these early lesions within the general population is unclear.

Prostate Cancer Risk

BRCA1/2 P/LP variants are associated with increase that BRCA prostate cancer. Carriers of a P/LP *BRCA1* variant have an estimated 7% to 26% cumulative lifetime risk of prostate cancer, while the cumulative lifetime risk is 19% to 61% for carriers of a P/LP *BRCA2* variant. There is evidence that advanced or metastatic prostate cancer is associated with carrying a *BRCA2* P/LP variant; it is not yet known if an aggressive phenotype is also associated with *BRCA1* P/LP variant. An international study including 5545 patients with prostate cancer (with European ancestry) showed that the frequency of a *BRCA2* P/LP variant was significantly higher in patients with aggressive disease (i.e., died from prostate cancer, metastatic disease, T4 disease, or T3 with Gleason score ≥ 8) than in patients with non-aggressive disease (OR, 3.19; 95% CI, 1.94–5.25). A study of a large cohort of patients from Spain with prostate cancer (N = 2019) showed that carriers of a P/LP *BRCA1/2* variant had significantly higher rates of aggressive prostate cancer (Gleason score ≥ 8), nodal involvement, and distant metastasis compared with non-carriers. In a sample of 692 patients with metastatic prostate cancer, unselected for family history or age at diagnosis, 5.3% carried a *BRCA2* P/LP variant, and 0.9% carried a *BRCA1* P/LP variant. In addition, analyses from a treatment center database showed that *BRCA1/2* and *ATM* (see below under *National Comprehensive Cancer Network®(NCCN®) Genetic Testing Criteria: Testing Criteria Related*

to Prostate Cancer) P/LP variant rates were highest in patients with metastatic disease (8.2%). This study also showed that carriers with prostate cancer had significantly decreased survival, compared with patients who were non-carriers (5 years vs. 16 years, respectively; $P < .001$). This association remained statistically significant when controlling for race, age, prostate-specific antigen (PSA), and Gleason score. Ashkenazi Jewish ancestry is also associated with *BRCA1/2* P/LP variants in patients with prostate cancer, with rates for *BRCA1* being 0% to 2%, and rates for *BRCA2* being 1% to 3%.

Pancreatic Cancer Risk

Prior to more widespread testing of individuals with pancreatic cancer for germline variants in cancer predisposition genes, studies showed that *BRCA1/2* P/LP variant rates in pancreatic cancer cases ranged from 1% to 11% for *BRCA1* and 0% to 17% for *BRCA2*. However, some of these studies included only patients with familial pancreatic cancer or those of Ashkenazi Jewish ancestry, both of whom may have a greater likelihood of testing positive for a *BRCA1/2* P/LP variant. More recent studies that used panel testing confirm that some pancreatic cancers harbor actionable *BRCA1/2* P/LP variants (0%–3% for *BRCA1* and 1%–6% for *BRCA2*). Cumulative lifetime risk of pancreatic cancer is as high as 3% for males with a *BRCA1/2* P/LP variant and 2.3% for females with a *BRCA1/2* P/LP variant. Patients with pancreatic cancer and Ashkenazi Jewish ancestry may have a greater likelihood of testing positive for a *BRCA1/2* P/LP variant, with prevalence of detected P/LP variants in this group ranging from 5.5% to 19%, with P/LP variants being more common for *BRCA2*.

Other Cancer and Health Risks

Some studies have suggested an increased risk specifically of serous uterine cancer in carriers of a P/LP *BRCA1/2* variant. Analyses from a multicenter perspective cohort study including 1083 carriers of a P/LP *BRCA1* variant who underwent RRSO without hysterectomy showed an increased risk for serous and/or serous-like endometrial cancer. A Dutch cohort study including 5980 carriers of a P/LP *BRCA1/2* variant showed a 2- to 3-fold increased risk for endometrial cancer, with the highest risks for serous-like (HR, 10.48; 95% CI, 2.95–37.20) and *p53*- abnormal endometrial cancer (HR, 15.71; 95% CI, 4.62–53.40) in carriers of a P/LP *BRCA1* variant. A systematic review and meta-analysis including 11 studies with 13,871 carriers of a P/LP *BRCA1/2* variant showed that the prevalence of endometrial cancer was 0.62% in carriers of a P/LP *BRCA1* variant and 0.47% in carriers of a P/LP *BRCA2* variant (relative RR, 1.18; 95% CI, 0.7–2.0). For uterine papillary serous carcinoma, the prevalence rates were 0.20% for *BRCA1* and 0.08% for *BRCA2* (relative RR, 1.39; 95% CI, 0.5–3.7). It has been suggested that the increased risk for endometrial cancer observed in some carriers of *BRCA1/2* P/LP variants may be due to the use of tamoxifen therapy by these patients rather than the presence of a P/LP variant.

A meta-analysis including five studies of patients with uterine serous cancer and Ashkenazi Jewish ancestry showed that *BRCA1/2* P/LP variant prevalence was greater in those with uterine serous cancer than in controls (also of Ashkenazi Jewish ancestry) (OR, 5.4; 95% CI, 2.2–13.1). In a retrospective case control study including 2627 Jewish Israeli carriers of a P/LP *BRCA1/2* variant (88% Ashkenazi Jewish), risk of developing uterine cancer was increased, with an observed-to-expected ratio of 3.98 (95% CI, 2.17–6.67; $P < .001$). This association persisted regardless of uterine cancer histology.

The absolute risk of uterine cancer in carriers of a P/LP *BRCA1/2* variant appears low overall, despite some evidence of increased risk. However, genetic testing, including for a P/LP *BRCA1* variant, may be considered for patients diagnosed with serous endometrial cancer.

Studies that investigated associations between *BRCA2* P/LP variant and cutaneous melanoma have drawn inconsistent conclusions, though there is some evidence of an association. One study showed that females carrying a P/LP *BRCA2* variant have an elevated risk for leukemia (standardized incidence ratio [SIR], 4.76; 95% CI, 1.21–12.96; $P = .03$), particularly females who have received chemotherapy (SIR, 8.11; 95% CI, 2.06–22.07; $P = .007$). Analyses of 3184 *BRCA1* and 2157 *BRCA2* families in the Consortium of Investigators of Modifiers of *BRCA1/2* showed that the cumulative lifetime risk of gastric cancer is 3.5% in both male and female carriers of a *BRCA2* pathogenic variant (compared to 1.6% and 0.7% in male and female carriers of a *BRCA1* pathogenic variant, respectively). A case-control analysis from Japan including 65,108 patients showed associations between gastric cancer and *BRCA1* (OR, 5.2; 95% CI, 2.6–10.5) and *BRCA2* (OR, 4.7; 95% CI, 3.1–7.1) P/L P variants, biliary tract cancer and P/LP *BRCA1* variant (OR, 17.4; 95% CI, 5.8–51.9), and esophageal cancer and P/LP *BRCA2* variant (OR, 5.6; 95% CI, 2.9–11.0). Finally, an analysis of 490 families with a known *BRCA1/2* P/LP variant showed an increased risk for ocular melanoma in carriers of a P/LP *BRCA2* variant (RR, 99.4; 95% CI, 11.1–359.8), though absolute risk is low.

In cases where both partners carry a P/LP *BRCA2* variant, there may be a high risk for the offspring to develop Fanconi anemia, a rare autosomal recessive condition. A review of 27 cases of Fanconi anemia with biallelic P/LP variants in *BRCA2* (FA-D1) showed a 97% cumulative risk of malignancy by age 5.2 years (79% risk of leukemia by age 10 years, 83% risk of any solid tumor by age 6.7 years, 85% risk of a brain tumor by age 9 years, and 63% risk of a Wilms tumor by age 6.7 years). Some case reports have also identified biallelic *BRCA1* P/LP variants causing Fanconi anemia-like disorder, particularly FANCS, a severe form of Fanconi anemia characterized by developmental delay, short stature, and microcephaly.

Risk Management

Recommendations for the medical management of *BRCA*-related cancers are based on an appreciation of the early onset and increased risk for associated cancers. An individual from a family with a known *BRCA1/2* P/LP variant who tests negative for the familial variant should be followed according to the recommendations for the general population for breast cancer (e.g., the NCCN Guidelines for Breast Cancer Screening and Diagnosis [available at www.NCCN.org]).

Breast Cancer Risk Management

Screening

Mammography has served as the standard screening modality for detection of breast cancer during the last few decades. There are currently no data indicating that mammography on its own reduces mortality in females with genetically increased risk for breast cancer. Also, false-negative mammography results are common and have been correlated with factors such as presence of a *BRCA1/2* P/LP variant and high breast tissue density, both of which may occur more frequently among younger females. Rapidly growing or aggressive breast tumors—also more common among younger females—have also been associated with decreased sensitivity of mammographic screening methods. Prospective studies on comparative surveillance modalities in females at high risk for familial breast cancer (i.e., confirmed or suspected *BRCA1/2* P/LP variant based on family history) have consistently reported higher sensitivity of MRI screening (77%–94%) compared with mammography (33%–59%) in detecting breast cancers. False-positive rates were higher with MRI in some reports, resulting in a slightly lower or similar specificity with MRI screening (81%–98%) compared with mammography (92%–100%). The sensitivity with ultrasound screening (33%–65%) appeared similar to that of mammography in this high-risk population. In a prospective screening trial (conducted from 1997–2009) that evaluated the performance of annual MRI and mammography in females (aged 25–65 years; N = 496) with confirmed P/LP *BRCA1/2* variant, sensitivity with MRI was significantly higher compared with mammography during the entire study period (86% vs. 19%; $P < .0001$). Factors such as age, P/LP variant type, or invasiveness of the tumor did not significantly influence the relative sensitivity of the two screening modalities. Importantly, the large majority (97%) of cancers detected by MRI screening were early-stage tumors. At a median follow-up of 8 years from diagnosis, none of the survivors (n = 24) had developed distant recurrence. In an analysis of 606 females with either a family history of breast cancer or who harbor a P/LP variant associated with increased risk for breast cancer, sensitivity of breast MRI screening was reported to be 79%, while specificity was reported to be 86%.

All of these studies discussed above evaluated a screening strategy that was conducted on an annual

basis, and many of the studies included individuals without known *BRCA1/2* P/LP variant status. A study of 1219 carriers of a P/LP *BRCA1* variant and 732 carriers of a P/LP *BRCA2* variant showed that the increased sensitivity of mammography in addition to MRI was greater for carriers of a P/LP *BRCA2* variant (12.6%) than for carriers of a P/LP *BRCA1* variant (3.9%). In a retrospective study, a different screening interval was evaluated, using alternating mammography and MRI screening every 6 months in females with a confirmed P/LP *BRCA1/2* variant (N = 73). After a median follow-up of 2 years, 13 breast cancers were detected among 11 females; 12 of the tumors were detected by MRI screening but not by mammography obtained 6 months earlier. The sensitivity and specificity with MRI screening was 92% and 87%, respectively.

The optimal surveillance approach in individuals assigned female at birth who are at high risk for familial breast cancer remains uncertain, especially for those between the ages of 25 and 30 years. While some studies have reported an association between radiation exposure from mammography and increased risk for breast cancer in carriers of a P/LP *BRCA1/2* variant, current data are insufficient to support risks of radiation. Nevertheless, one of the potential benefits of incorporating MRI modalities into surveillance strategies may include minimizing the radiation risks associated with mammography, in addition to the higher sensitivity of MRI screening in detecting tumors. The use of MRI, however, may potentially be associated with higher false-positive results and higher costs relative to mammography. The combined use of digital mammography (two-dimensional, 2D) in conjunction with digital breast tomosynthesis (DBT) appears to improve cancer detection and reduce false-positive call back rates. Tomosynthesis allows acquisition of three-dimensional (3D) data using a moving x-ray and digital detector. These data are reconstructed using computer algorithms to generate thin sections of images. The combined use of 2D and DBT results in double the radiation exposure compared with mammography alone. However, this increase in radiation dose falls below dose limits of radiation set by the U.S. Food and Drug Administration (FDA) for standard mammography. The radiation dose can be minimized by newer tomosynthesis techniques that create a synthetic 2D image, which may obviate the need for a conventional digital image. In carriers of a *BRCA1/2* P/LP variant who are <30 years of age, breast MRI screening is preferred over mammography due to lack of data to support benefit due to less sensitivity for detection of tumors associated with mammography. Studies have reported that deposits of gadolinium, a component of MRI contrast agents, remain in the brain of some patients who undergo 4 or more contrast MRI scans, long after the last administration. Retention of gadolinium has also been seen in the bone. In 2017, the FDA issued an update stating that its review of available data had not identified adverse health effects from gadolinium retained in the brain and that patients should read a medication guide prior to receiving gadolinium. However, review of the evidence will continue.

The appropriate imaging modalities and surveillance intervals are still under investigation. In a report

based on a computer simulation model that evaluated different annual screening strategies in carriers of a P/LP *BRCA1/2* variant, a screening approach that included annual MRI starting at 25 years of age combined with alternating digital mammography/MRI starting at 30 years of age was shown to be the most effective strategy when radiation risks, life expectancy, and false-positive rates were considered. Future prospective trials are needed to evaluate the different surveillance strategies in individuals at high risk for familial breast cancer. For an individual assigned female at birth who is a carrier of a *BRCA1/2* P/LP variant, training in breast awareness with regular monthly practice should begin at 18 years of age, and clinical breast examinations should be conducted every 6 to 12 months, beginning at 25 years of age. Between the ages of 25 and 29 years, these individuals should have annual breast MRI screening with contrast (to be performed on days 7 to 15 of menstrual cycle for premenopausal individuals) or annual mammograms only if MRI is not available. The age to begin screening can be individualized if the family history includes a breast diagnosis prior to 30 years of age. Breast MRI screening is preferred over mammogram in the 25- to 29-year age group. High-quality breast MRI screening should consist of the following: dedicated breast coil, ability to perform biopsy under MRI guidance, experienced radiologists in breast MRI, and regional availability. Between 30 and 75 years of age, annual mammogram and breast MRI with contrast should both be done. After 75 years of age, management should be considered on an individual basis. In females treated for breast cancer who have not had bilateral mastectomy, mammography and breast MRI screening with contrast should continue as recommended based on age. Emerging evidence suggests that abbreviated-protocol breast MRI is a screening strategy that warrants further investigation in carriers of a *BRCA1/2* P/LP variant.

Carriers of a *BRCA1/2* P/LP variant who were assigned male at birth should have an annual clinical breast examination and undergo training in breast self-examination with regular monthly practice starting at 35 years of age. A 12-year longitudinal observational study evaluated the outcomes of mammography screening in 1869 males who were at increased risk of developing breast cancer (i.e., personal or family history of breast cancer and/or germline P/LP variant associated with breast cancer, mostly *BRCA1* and *BRCA2*). Node-negative breast cancer was identified in five males (18 per 1000 examinations), which is greater than the cancer detection rates in both average-risk and high-risk females who undergo breast screening. Harboring a P/LP variant ($n = 47$) was associated with breast cancer (OR, 7; 95% CI, 2–29; $P = .006$). Because of the lack of screening, males diagnosed with breast cancer have historically presented with advanced stage disease. Annual mammogram in males may be considered, especially in those carrying a *BRCA2* P/LP variant, beginning at age 50 or 10 years before the earliest known male breast cancer in the family (whichever comes first). Though gynecomastia may be associated with breast cancer, it is not a risk factor for breast cancer. As per the American College of Radiology (ACR) Appropriateness Criteria, gynecomastia does not need to be present in order to obtain a diagnostic mammogram.

Bilateral Total Mastectomy

Two meta-analyses show that prophylactic bilateral mastectomy reduces the risk for breast cancer. Only one of these analyses showed that risk-reducing surgery is significantly associated with reduced mortality. Retrospective studies and small prospective studies provide support for concluding that RRM provides a high degree of protection against breast cancer in females carrying a P/LP *BRCA1/2* variant.

It is important that the potential psychosocial effects of RRM are addressed. A 2018 Cochrane review including 20 studies that evaluated psychosocial effects of RRM showed that patients are generally satisfied with their decision, with reported decreases in worry about breast cancer, but negative impacts on body image and sexuality have also been reported. Additional research is needed to further evaluate the psychosocial impact of RRM. RRM is also associated with long-term physical symptoms, such as lower sensitivity to touch, pain, tingling, infection, and edema. Multidisciplinary consultations are recommended prior to surgery and should include the discussions of the risks and benefits of surgery, and surgical breast reconstruction options. Immediate breast reconstruction is an option following RRM, and early consultation with a reconstructive surgeon is recommended for those considering either immediate or delayed breast reconstruction. Nipple-sparing mastectomy has been suggested to be a safe and effective risk reduction strategy for patients carrying a *BRCA1/2* P/LP variant, although more data and longer follow-up are needed.

The NCCN Guidelines Panel supports discussion of the option of RRM for individuals assigned female at birth on a case-by-case basis. Counseling for this risk-reducing surgery should include discussion of extent of cancer risk reduction/protection, risks associated with surgeries, breast reconstructive options, and management of menopausal symptoms. Since risk of breast cancer remains increased with age in carriers of a *BRCA1/2* P/LP variant, age and life expectancy should also be considered during this counseling, as well as family history. It is important to address the psychosocial and quality-of-life aspects of undergoing risk-reducing surgical procedures.

Chemoprevention

The use of selective estrogen receptor modulators (i.e., tamoxifen, raloxifene) has been shown to reduce the risk for invasive breast cancer in individuals considered at high risk for developing breast cancer, especially ER-positive disease. However, only limited data are available on the specific use of these agents for primary prevention in patients with *BRCA1/2* P/LP variants. As previously discussed, patients with *BRCA1/2* P/LP variants who are diagnosed with breast cancer have elevated risks for

developing contralateral breast tumors. In one of the largest prospective series of carriers of a P/LP *BRCA1/2* variant evaluated, the mean cumulative lifetime risks for contralateral breast cancer were estimated to be 83% for carriers of a P/LP *BRCA1* variant and 62% for carriers of a P/LP *BRCA2* variant. Patients carrying a P/LP *BRCA1/2* variant who have intact contralateral breast tissue (and who do not undergo oophorectomy or receive chemoprevention) have an estimated 40% risk for contralateral breast cancer at 10 years, though risk is dependent on age of first breast cancer diagnosis. Case-control studies from the Hereditary Breast Cancer Clinical Study Group reported that the use of tamoxifen protected against contralateral breast cancer with an OR of 0.38 (95% CI, 0.19–0.74) to 0.50 (95% CI, 0.30–0.85) among carriers of a P/LP *BRCA1* variant and 0.42 (95% CI, 0.17–1.02) to 0.63 (95% CI, 0.20–1.50) among carriers of a P/LP *BRCA2* variant. This translates to an approximately 45% to 60% risk reduction for contralateral tumors among carriers of a P/LP *BRCA1/2* variant with breast cancer. The data were not consistent in regard to the protective effects of tamoxifen in the subset of carriers of a P/LP *BRCA1/2* variant who also underwent oophorectomy. In addition, no data were available on the estrogen receptor status of the tumors. An evaluation of the subset of healthy carriers of a P/LP *BRCA1/2* variant in the Breast Cancer Prevention Trial revealed that breast cancer risk was reduced by 62% in carriers of a P/LP *BRCA2* variant receiving tamoxifen relative to placebo (risk ratio, 0.38; 95% CI, 0.06–1.56). However, an analysis of females who developed breast cancer during their participation in this trial showed that tamoxifen use was not associated with a reduction in breast cancer risk in carriers of a P/LP *BRCA1* variant. These findings may be related to the greater likelihood for development of estrogen receptor-negative tumors in carriers of a P/LP *BRCA1* variant relative to carriers of a P/LP *BRCA2* variant. However, this analysis was limited by the very small number of individuals with a P/LP *BRCA1/2* variant ($n = 19$; 7% of participants diagnosed with breast cancer). Common single-nucleotide polymorphisms have been identified in genes (*ZNF423* and *CTSO*) that are involved in estrogen-dependent regulation of *BRCA1* expression. These gene variants were associated with alterations in breast cancer risk during treatment with selective estrogen receptor modulators and may eventually pave the way for predicting the likelihood of benefit with these chemo preventive approaches in individual patients.

The aromatase inhibitors (AIs) exemestane and anastrozole have been demonstrated to be effective in preventing breast cancer in postmenopausal individuals considered to be at high risk of developing breast cancer. However, to date, there is little evidence supporting the use of AIs as an effective chemo preventive approach for individuals with a *BRCA1/2* P/LP variant. A retrospective study showed that AIs may reduce the risk of contralateral breast cancer in females with a *BRCA1/2* P/LP variant and ER-positive breast cancer who take AIs as adjuvant therapy, but these data are currently published in abstract form only.

Studies on the effect of oral contraceptive use on breast cancer risk among carriers of a P/LP *BRCA1/2* variant have reported conflicting data. In one case-control study, use of oral contraceptives was associated with a modest but statistically significant increase in breast cancer risk among carriers of a P/LP *BRCA1* variant (OR, 1.20; 95% CI, 1.02–1.40), with breast cancer risk in these carriers being associated with 5 or more years of oral contraceptive use (OR, 1.33; 95% CI, 1.11–1.60), breast cancer diagnosed before 40 years of age (OR, 1.38; 95% CI, 1.11–1.72), and use of oral contraceptives before 1975 (OR, 1.42; 95% CI, 1.17–1.75). Oral contraceptive use was not significantly associated with breast cancer in carriers of a *BRCA2* P/LP variant in this study. In another case-control study, use of oral contraceptives for at least 5 years was associated with a significantly increased risk for breast cancer in carriers of a P/LP *BRCA2* variant (OR, 2.06; 95% CI, 1.08–3.94); results were similar when only the cases with oral contraceptive use on or after 1975 were considered. Oral contraceptive use for at least 1 year was not significantly associated with breast cancer risk in carriers of a P/LP *BRCA1* or *BRCA2* variant in this study. In a third case-control study, the use of low-dose oral contraceptives for at least 1 year was associated with significantly decreased risks for breast cancer among carriers of a P/LP *BRCA1* variant (OR, 0.22; 95% CI, 0.10–0.49; $P < .001$), though not for carriers of a P/LP *BRCA2* variant. A case control study found an increase in breast cancer risk for *BRCA1* mutation carriers who started oral contraception before the age of 20 years (OR, 1.45; 95% CI, 1.20–1.75; $P = .0001$). Associations after age 20 were not found to be statistically significant.

To summarize, findings from case-control studies are inconsistent regarding the effect of oral contraceptive use on the association of P/LP *BRCA1* or *BRCA2* variant and breast cancer risk. Oral contraceptive use for more than 5 years may be associated with increased risk. Findings from meta-analyses are also conflicting, with several showing that oral contraceptive use is not significantly associated with breast cancer risk in carriers of a P/LP *BRCA1/2* variant, while others show mixed results based on subgroup analysis, study type, and population. Another 2022 meta-analysis including 12 studies showed that the use of oral contraceptives was associated with an increased risk of breast cancer in carriers of both *BRCA1* and *BRCA2* P/LP variants, particularly in those who used oral contraceptives for 5 years or longer. A study examining a hypothetical cohort of 10,000 females found that the use of combined oral contraceptives was associated with increased breast cancer risk in those with a P/LP *BRCA1* variant, assuming 10 years of continuous oral contraceptive use. The breast cancer risk difference attributable to oral contraceptive use increased throughout life for carriers of a P/LP *BRCA1/2* variant (compared to ovarian and endometrial cancers, which showed decreasing incidences as age increased).

Differences in the study design used by these case-control studies make it difficult to compare outcomes between studies, and likely account for the conflicting results. The design of these studies

might have differed with regard to factors such as the criteria for defining the “control” population for the study (e.g., non-*BRCA1/2* carriers vs. P/LP variant carriers without a cancer diagnosis), consideration of family history of breast or ovarian cancer, baseline demographics of the population studied (e.g., nationality, ethnicity, geographic region, age groups), age of onset of breast cancer, and formulations or duration of oral contraceptives used.

Ovarian/Uterine Cancer Risk Management

Bilateral Salpingo-Oophorectomy

Carriers of a confirmed *BRCA1/2* P/LP variant are at increased risk for both breast and ovarian cancers (including fallopian tube cancer and primary peritoneal cancer). Although the risk for ovarian cancer is generally considered to be lower than the risk for breast cancer in carriers of a P/LP *BRCA1/2* variant, the absence of reliable methods of early detection and the poor prognosis associated with advanced ovarian cancer have lent support for the performance of bilateral RRSO after completion of childbearing.

A 2014 observational prospective study of 5783 females carrying a P/LP *BRCA1/2* variant showed that ovarian cancer is more prevalent in individuals carrying a *BRCA1* (4.2%) P/LP variant than those carrying a *BRCA2* (0.6%) P/LP variant. In carriers of a P/LP *BRCA1* variant, prevalence of ovarian, fallopian tube, and peritoneal cancers found during risk-reducing surgery was 1.5% for those <40 years of age and 3.8% in those between the ages of 40 and 49 years. The highest incidence rate for carriers of a P/LP *BRCA1* variant was observed between the ages of 50 and 59 years (annual risk, 1.7%); for carriers of a P/LP *BRCA2* variant, the highest incidence rate was observed between the ages of 60 and 69 years (annual risk, 0.6%). A more recent retrospective cohort study including 474 carriers of a P/LP *BRCA1/2* variant who were diagnosed with high-grade serous ovarian cancer showed that age of diagnosis was significantly greater in carriers of a P/LP *BRCA2* variant, compared to carriers of a P/LP *BRCA1* variant (58.4 vs. 53.3 years; $P = .001$). Therefore, the recommended age for RRSO should be younger for carriers of a P/LP *BRCA1* variant than for carriers of a P/LP *BRCA2* variant.

The effectiveness of RRSO in reducing the risk for ovarian cancer in carriers of a *BRCA1/2* P/LP variant has been demonstrated in a number of studies. For example, results of a meta-analysis involving 10 studies of carriers of a *BRCA1/2* P/LP variant showed an approximately 80% reduction in the risk for ovarian or fallopian tube cancer following RRSO. In a large prospective study of females who carried deleterious *BRCA1/2* variants ($N = 1079$), RRSO significantly reduced the risk for *BRCA1*- associated gynecologic tumors (including ovarian, fallopian tube, or primary peritoneal cancers) by 85% compared with observation during a 3- year follow-up period (HR, 0.15; 95% CI, 0.04–0.56; $P = .005$). An

observational study of 5783 females carrying a P/LP *BRCA1/2* variant showed that risk-reducing oophorectomy reduces risk for ovarian, fallopian tube, or peritoneal cancer by 80% (HR, 0.20; 95% CI, 0.13–0.30) and all-cause mortality by 77% (HR, 0.23; 95% CI, 0.13–0.39). RRSO reduces mortality at all ages in carriers of a P/LP *BRCA1* variant, but among carriers of a P/LP *BRCA2* variant, RRSO is only associated with reduced mortality in those between the ages of 41 and 60 years.

A 1% to 4.3% residual risk for a primary peritoneal carcinoma has been reported in some studies. An analysis of 36 carriers of a *BRCA1/2* P/LP variant who developed peritoneal carcinomatosis following RRSO showed that 86% were carriers of a *BRCA1* P/LP variant specifically. When comparing to 113 carriers of a P/LP *BRCA1/2* variant who did not develop peritoneal carcinomatosis following RRSO, females who eventually developed peritoneal carcinomatosis were older at time of RRSO ($P = .025$) and had a greater percentage of serous tubal intraepithelial carcinoma (STIC) in their RRSO specimen ($P < .001$), supporting the removal of the fallopian tubes as part of the risk-reducing procedure. A systematic review and individual patient data meta-analysis including 17 studies and 3121 patients showed that STIC at RRSO was strongly associated with increased risk of peritoneal carcinomatosis (HR, 33.9; 95% CI, 15.6–73.9; $P < .001$). Further, an analysis from a multicenter prospective cohort study ($N = 1083$) showed an increased risk for serous and/or serous-like endometrial cancer in females carrying a P/LP *BRCA1* variant who underwent RRSO without hysterectomy.

RRSO may provide an opportunity for gynecologic cancer detection in carriers of a P/LP *BRCA1/2* variant. An analysis of 966 RRSO procedures showed that invasive or intraepithelial ovarian, tubal, or peritoneal neoplasms were detected in 4.6% of carriers of a P/LP *BRCA1* variant and 3.5% of carriers of a P/LP *BRCA2* variant. Carrying a *BRCA1/2* P/LP variant was associated with detection of clinically occult neoplasms during RRSO ($P = .006$). Another study including 2557 asymptomatic carriers of P/LP *BRCA1/2* variant enrolled in the Hereditary Breast and Ovarian Cancer in the Netherlands Study showed that high-grade serous carcinoma was detected in 1.5% of carriers of a P/LP *BRCA1* variant and in 0.6% of carriers of a P/LP *BRCA2* variant at time of RRSO. The fallopian tubes were the primary location of cancer in 73.3% of carriers for whom cancer was detected.

In early studies, RRSO was reported to reduce the risk for breast cancer in carriers of a P/LP *BRCA1/2* variant. In the case-control international study by Eisen et al, a 56% (OR, 0.44; 95% CI, 0.29–0.66; $P < .001$) and a 43% (OR, 0.57; 95% CI, 0.28–1.15; $P = .11$) breast cancer risk reduction (adjusted for oral contraceptive use and parity) were reported following RRSO in carriers of a *BRCA1* and a *BRCA2* P/LP variant, respectively. A study comparing breast cancer risk in females carrying a P/LP *BRCA1/2* variant who had undergone RRSO with carriers of these P/LP variants who opted for surveillance only also

showed reduced breast cancer risk in females who underwent RRSO (HR, 0.47; 95% CI, 0.29–0.77). These studies were further supported by a meta-analysis that found similar reductions in breast cancer risk of approximately 50% for carriers of a P/LP *BRCA1/2* variant following RRSO.

Results of a prospective cohort study suggested that RRSO may be associated with a greater reduction in breast cancer risk for carriers of a P/LP *BRCA2* variant compared with carriers of a *BRCA1* P/LP variant. Another retrospective analysis including 676 females with stage I or II breast cancer and a P/LP *BRCA1/2* variant showed that oophorectomy was associated with decreased risk of mortality from breast cancer in carriers of a P/LP *BRCA1* variant (HR, 0.38; 95% CI, 0.19–0.77; $P = .007$), but not in carriers of a P/LP *BRCA2* variant ($P = .23$).

The reduction in breast cancer risk following RRSO was questioned in a prospective cohort study from the Netherlands ($N = 822$), which did not find a statistically significant difference in breast cancer incidence between carriers of a *BRCA1/2* P/LP variant who opted for an RRSO and females who did not, regardless of whether the P/LP variant was *BRCA1* or *BRCA2*. Study investigators argued that previous study findings showing a 50% decrease in breast cancer risk may have been influenced by bias, specifically inclusion of patients with a history of breast or ovarian cancer in the comparison group and immortal person-time bias. One study that corrected for immortal person-time bias as a result of this analysis continued to find a protective effect of RRSO on breast cancer incidence in carriers of a P/LP *BRCA1/2* variant (HR, 0.59; 95% CI, 0.42–0.82; $P < .001$). Another prospective cohort analysis including 1289 carriers of a P/LP *BRCA1/2* variant unaffected with breast cancer (196 eventually being diagnosed) also showed that, when RRSO was treated as a time-dependent variable, it was no longer associated with breast cancer risk. A meta-analysis including 19 studies of the association between RRSO, and breast cancer risk and mortality showed a protective effect in studies published earlier than 2016, but not in studies published in 2016 or later ($n = 3$).

Results from one of the earlier studies showed that greater reductions in breast cancer risk were observed in females carrying a P/LP *BRCA1* variant who had an RRSO at ≤ 40 years of age (OR, 0.36; 95% CI, 0.20–0.64) relative to carriers of a P/LP *BRCA1* variant aged 41 to 50 years who had this procedure (OR, 0.50; 95% CI, 0.27–0.92). A nonsignificant reduction in breast cancer risk was found for females aged ≥ 51 years, although only a small number of females were included in this group. However, results from another early study also suggested that RRSO after 50 years of age is not associated with a substantial decrease in breast cancer risk. A 2017 study showed that oophorectomy was not significantly associated with decreased risk of breast cancer in carriers of a P/LP *BRCA1/2* variant ($N = 3722$). However, stratified analyses in carriers of a P/LP *BRCA2* variant who were

diagnosed with breast cancer before 50 years of age showed that oophorectomy was associated with an 82% reduction in breast cancer (HR, 0.18; 95% CI, 0.05–0.63; $P = .007$). The risk reduction in carriers of a P/LP *BRCA1* variant was not statistically significant ($P = .51$). A 2020 study including 853 premenopausal carriers of a P/LP *BRCA1/2* variant showed the opposite: that premenopausal RRSO decreased breast cancer risk in carriers of a *BRCA1* P/LP variant (HR, 0.45; 95% CI, 0.22–0.92), but not in carriers of a *BRCA2* P/LP variant (HR, 0.77; 95% CI, 0.35–1.67). Analysis for this study began observation 6 months after genetic testing to avoid event-free time bias.

A large case series published in 2021 addressed the permanent exposure hypothesis that has potentially dampened the strength of the conclusions drawn from previous studies on the association between RRSO and breast cancer risk reduction. Specifically, some of these earlier studies assumed that this association remains constant each year following RRSO. This study, which included 876 families with a known *BRCA1* or *BRCA2* P/LP variant, showed that RRSO reduced risk of breast cancer within 5 years following the surgery (HR, 0.28; 95% CI, 0.10–0.63 and HR, 0.19; 95% CI, 0.06–0.71, respectively). More than 5 years after RRSO, breast cancer risk reduction diminished but continued to be significant for carriers of a *BRCA1* P/LP variant (HR, 0.64; 95% CI, 0.38–0.97), while the reduction was no longer statistically significant for carriers of a *BRCA2* P/LP variant (HR, 0.99; 95% CI, 0.84–1.00).

To summarize, studies suggest a benefit of RRSO on breast cancer risk, but the magnitude of the effect based on age remains uncertain.

Two systematic reviews showed that HRT does not negate the reduction in breast cancer risk associated with the surgery. One of these reviews showed that breast cancer risk tended to be lower in females who received estrogen only, compared to estrogen plus progesterone (OR, 0.62; 95% CI, 0.29–1.31). It is important to have a discussion about the potential risks and benefits of HRT in carriers of a P/LP variant following RRSO, given the limitations inherent in nonrandomized studies.

Salpingectomy (surgical removal of the fallopian tube with delayed oophorectomy) reduces the risk of ovarian cancer in the general population and is an option for premenopausal patients with hereditary cancer risk who are not yet ready for oophorectomy in the context of a clinical trial. Salpingectomy is currently not proven to improve outcomes and continues to be a procedure still under investigation. CA125 and pelvic ultrasound are recommended for preoperative planning. Continuation of combination oral contraceptives or hormonal intrauterine device (IUD) for continued ovarian cancer risk reduction while ovaries remain in place may be considered. Clinical trials of interval salpingectomy with delayed oophorectomy are ongoing (e.g., NCT02321228, NCT01907789, NCT04294927).

Some studies suggest a link between BRCA P/LP variants and development of serous uterine cancer (primarily with BRCA1), although the overall risk for uterine cancer was not increased when controlling for tamoxifen use. Individuals who undergo hysterectomy at the time of RRSO are candidates for estrogen-alone HRT, which is associated with a decreased risk of breast cancer, compared to combined estrogen and progesterone, which is required when the uterus is left in situ.^{331,332,338} Risk of pelvic floor dysfunction or urinary incontinence after hysterectomy is influenced by factors other than hysterectomy alone. Long-term follow-up studies indicate that the risks are <5% if there is no preceding pelvic organ prolapse. For patients who choose to undergo RRSO, the provider may discuss the risks and benefits of concurrent hysterectomy, but more data are needed to determine the magnitude of the association between BRCA variants and development of serous uterine cancer.

HRT is generally not contraindicated and thus should be discussed with premenopausal patients who do not have a personal history of breast cancer. HRT recommendations should be tailored depending on each patient's personal history of breast cancer and/or breast cancer risk reduction strategies. In patients for whom the uterus is left in place at time of RRSO, there are several hormone replacement options. There is evidence that levonogestrel IUD is associated with lower risk of breast cancer compared to risk of breast cancer from orally administered progestin. However, combination estrogen/progestin HRT continues to be an option after RRSO, though counseling should include bleeding precautions and uterine cancer risk awareness. Combined estrogen with a selective estrogen receptor modulator (e.g., bazedoxifene) is also an option.

The NCCN Guidelines Panel recommends RRSO for carriers of a known *BRCA1/2* P/LP variant, typically between 35 and 40 years of age for carriers of a *BRCA1* P/LP variant. Since ovarian cancer onset tends to be later in carriers of a *BRCA2* P/LP variant, it is reasonable to delay RRSO for management of ovarian cancer risk until between 40 and 45 years of age, unless age at diagnosis in the family warrants earlier age for consideration of this prophylactic surgery. Peritoneal washings should be performed at surgery, and pathologic assessment should include fine sectioning of the ovaries and fallopian tubes. The protocol published by CAP (2009) can be consulted for details on specimen evaluation. See the NCCN Guidelines for Ovarian Cancer for treatment of findings (available at www.NCCN.org).

The decision to undergo RRSO is a complex one and should be made ideally in consultation with a gynecologic oncologist, especially when the patient wishes to undergo RRSO before the age at which it is typically recommended. Topics that should be addressed include impact on reproduction, impact on breast and ovarian cancer risk, risks associated with premature menopause (e.g., osteoporosis, cardiovascular disease, cognitive changes, changes to vasomotor symptoms, sexual concerns), and other medical issues. Preoperative menopause management consultation may be considered for

patients who are premenopausal at time of RRSO.

Chemoprevention

With respect to the evidence regarding the effect of oral contraceptives on cancer risks in carriers of a known *BRCA1/2* P/LP variant, case-control studies have demonstrated that oral contraceptives reduced the risk for ovarian cancer by 45% to 50% in carriers of a P/LP *BRCA1* variant and by 60% in carriers of a P/LP *BRCA2* variant. Moreover, risks appeared to decrease with longer duration of oral contraceptive use. An analysis of four meta-analyses, one review, one case-control study, and one retrospective cohort study showed that oral contraceptive use was associated with reduced risk of ovarian cancer in carriers of a known *BRCA1/2* P/LP variant. This review also showed that longer duration of use was associated with decreased ovarian cancer risk. A modeling study including hypothetical cohorts of 10,000 females with a P/LP *BRCA1* variant and 10,000 females with a P/LP *BRCA2* variant showed that combined oral contraceptive use was associated with short-term increased risk of breast cancer but decreased long-term risk of ovarian cancer and endometrial cancer (regardless of variant). However, this study found that the long-term benefit was reduced following menopausal HRT after bilateral salpingo-oophorectomy. Oral contraceptives may be considered for the purpose of ovulation suppression and are not contraindicated for birth control purposes.

Screening

Studies assessing whether ovarian cancer screening procedures are sufficiently sensitive or specific have yielded mixed results. The UK Collaborative Trial of Ovarian Cancer Screening (UKCTOCS), which assessed multimodality screening with transvaginal ultrasound (TVUS) and CA-125 versus either TVUS alone or no screening, showed that multimodality screening is more effective at detecting early-stage cancer; however, after a median of 11 years of follow-up, a significant mortality reduction was not observed. In phase II of the UK Familial Ovarian Cancer Screening Study (UK FOCSS), 4348 females with an estimated lifetime ovarian cancer risk no less than 10% underwent ovarian cancer screening via serum CA-125 tests every 4 months (with the risk of ovarian cancer algorithm [ROCA] used to interpret results) and TVUS (annually or within 2 months if abnormal ROCA score). Thirteen patients were diagnosed with ovarian cancer as a result of the screening protocol, with 5 of the 13 patients being diagnosed with early-stage cancer. Sensitivity, positive predictive value, and negative predictive value of the screening protocol for detecting ovarian cancer within 1 year were 94.7%, 10.8%, and 100%, respectively. A third study including 3692 females who were at increased familial/genetic risk of ovarian cancer (i.e., known P/LP *BRCA1/2* variant in the family and/or family history of multiple breast and/or ovarian cancers) showed that a ROCA-based screening protocol (i.e., serum CA-125 testing every 3 months with annual TVUS annually or sooner depending on CA-125 test results) identified 6 incidental ovarian cancers, of which 50% were early stage. The results of these studies suggest a potential stage shift when a ROCA-based ovarian cancer screening protocol is

followed in high-risk females, though it remains unknown whether this screening protocol impacts survival. TVUS and serum CA-125 screening are recommended for preoperative planning.

Risk Management for Other Cancers

Screening for prostate cancer starting at 40 years of age is recommended for carriers of a P/LP *BRCA2* variant and should be considered for carriers of a P/LP *BRCA1* variant. See the NCCN Guidelines for Prostate Cancer Early Detection (available at www.NCCN.org). General melanoma risk management is also indicated, such as annual full body skin exam and minimizing ultraviolet (UV) exposure. There are no specific screening guidelines for melanoma, though more information can be found at the website for the Skin Cancer Foundation (www.skincancer.org). Information on pancreas screening can be found below under *Hereditary Pancreatic Cancer*.

Other P/LP Variants Associated with Breast/Ovarian Cancer

Prior to 2020, the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, and Pancreatic focused largely on testing criteria for *BRCA1/2*, *PTEN*, and *TP53* and appropriate risk management for carriers of these P/LP variants. There is now strong evidence that genes beyond *BRCA1/2* confer markedly increased risk of breast and/or ovarian cancers. These genes include *ATM*, *BARD1*, *BRIP1*, *CDH1*, *CHEK2*, *MSH2*, *MSH6*, *MLH1*, *PMS2*, *EPCAM*, *NF1*, *PALB2*, *RAD51C*, *RAD51D*, and *STK11*. The panel's recommendations for cancer risk management intervention for carriers of P/LP variants associated with breast and/or ovarian cancer risk are based on absolute lifetime risk estimates. Cancer risk management intervention may be recommended when a carrier's absolute risk exceeds that of the average-risk population (i.e., 12%–13% for breast cancer and 1%–2% for ovarian cancer, based on SEER registry data). Strength of the evidence supporting risk estimates should also be evaluated when determining appropriate risk management for carriers of a P/LP variant. For example, prospective cohort studies in a population-based setting can be considered very strong evidence, while limited conclusions can be drawn from case series or studies with small samples.

The age at which breast screening is recommended may be impacted by the presence of risk factors such as family history of breast cancer, especially early-onset breast cancer. In those with a family history of early-onset breast cancer, breast screening may begin 5 to 10 years earlier than the youngest breast cancer diagnosis in the family. In individuals assigned female at birth treated for breast cancer who have not had bilateral mastectomy, breast screening should continue as recommended based on age. Currently there is insufficient evidence to recommend RRM in carriers of moderately penetrant

P/LP variants, though this option may be considered and discussed in the presence of a family history of breast cancer. Absolute risk estimates for breast cancer provided in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, and Pancreatic are for primary breast cancer unless otherwise noted. With the exception of *BRCA2*, risk of hereditary contralateral breast cancer risk decreases once postmenopausal and is equivalent to sporadic breast cancer risk after age.

RRSO may be considered when risk of developing ovarian cancer exceeds that of the average-risk population. The panel uses a threshold of 5% for a recommendation to discuss RRSO. For P/LP variants for which lifetime risk estimates are <5% but greater than population risk (e.g., *PALB2*), RRSO may be considered based on family history. The decision to carry out RRSO should not be made lightly, given the impact of premature menopause. RRSO is recommended for ovarian cancer risk management in carriers of a P/LP variant in an ovarian cancer susceptibility gene. However, some may choose to not receive an RRSO.

The P/LP variants described below may be included concurrently in panel testing (see *Multi-Gene Testing* above). Lower penetrance genes that may be included as part of multi-gene testing but for which there is currently insufficient evidence of an association with breast and/or ovarian cancer include: *FANCC*, *MRE11A*, *MUTYH* heterozygotes, *NBN*, *RECQL4*, *RAD50*, *RINT1*, *SLX4*, *SMARCA4*, and *XRCC2*. Risk management recommendations for these genes should take into account family history and other clinical factors. A more comprehensive review of these lower- penetrance genes is described in another publication.

ATM

P/LP variants in the *ATM* (ataxia-telangiectasia mutated) gene may increase the risk for breast cancer. A meta-analysis including 19 studies showed that the cumulative lifetime risk for primary breast cancer in individuals with an *ATM* P/LP variant is 6% by age 50 years and 33% by age 80 years. An analysis of 251 females who tested positive for ≥ 1 P/LP variant in a breast cancer susceptibility gene showed that the cumulative lifetime risk for primary breast cancer in individuals with an *ATM* P/LP variant is 31.2% by age 70 years. A comparative modeling analysis published in 2022 showed that the mean model-estimated lifetime risk of developing primary breast cancer was 20.9% (95% CI, 23.4%–31.7%) in females who carry an *ATM* P/LP variant. A meta-analysis of three cohort studies of relatives with ataxia-telangiectasia showed an estimated RR of 2.8 (90% CI, 2.2–3.7; $P < .001$). Other analyses of patients with breast cancer showed that about 1% had an *ATM* P/LP variant. Studies to date suggest lifetime risk of developing primary breast cancer in females who carry an *ATM* P/LP variant in the range of 20% to 30%. Ten-year cumulative risk of developing contralateral breast cancer in females

who carry an *ATM* P/LP variant is 4%, though this estimate is based on 7 cases. Therefore, additional studies are needed to confirm and refine this estimate.

The association between specific types of *ATM* genetic variants and breast cancer susceptibility is less clear, with some evidence showing that certain missense P/LP variants may act in a dominant-negative fashion to increase cancer risk, relative to truncating P/LP variants. A meta-analysis including five studies showed that carriers of an *ATM* P/LP variant have a 38% lifetime risk of developing primary breast cancer, with carriers of the c.7271T>G missense P/LP variant having a 69% risk of developing primary breast cancer by 70 years of age. An analysis from a case-control study (42,671 breast cancer cases and 42,164 controls) showed a significant association between the c.7271T>G variant and primary breast cancer risk (OR, 11.60; 95% CI, 1.50–89.90; $P = .001$). An analysis of 27 families in which P/LP *ATM* variants were identified showed an association between the c.7271T>G variant and increased risk for primary breast cancer (HR, 8.0; 95% CI, 2.3–27.4; $P < .001$).

The 2022 comparative modeling analysis by Lowry et al showed that beginning annual MRI screening at age 30 to 35 years may reduce breast cancer mortality by more than 55% in carriers with a P/LP *ATM* variant. The panel also recommends annual mammogram for carriers with a P/LP *ATM* variant beginning at 40 years of age. Age at which to initiate MRI in carriers with a P/LP *ATM* variant depends on a number of risk factors, including family history, age, breast density, and patient preference. There are no data on the benefit of RRM for carriers of a P/LP *ATM* variant, but this procedure may be considered based on family history. Results of the case-control WECARE study suggested that radiation exposure may be associated with increased risk for contralateral breast cancer in females who are carriers of very rare *ATM* missense variants. However, these variants are not P/LP, and a meta-analysis including five studies showed that radiation therapy (with conventional dosing) is not contraindicated in patients with a heterozygous *ATM* P/LP variant. Therefore, radiation therapy does not need to be avoided in these carriers who are diagnosed with cancer.

Large studies of patients with ovarian cancer have shown that there may be a slightly increased risk for ovarian cancer in carriers of an *ATM* P/LP variant, but there is currently insufficient evidence to recommend RRSO in these carriers. Given the association between *ATM* and development of the autosomal recessive condition ataxia telangiectasia, counseling for carriers of *ATM* P/LP variants should include a discussion of reproductive options. *ATM* P/LP variants have been found in patients with pancreatic cancer, with a lifetime risk of about 5% to 10% (see *Hereditary Pancreatic Cancer*, below). There is emerging evidence that *ATM* P/LP variants are associated with increased risk for prostate

cancer. Prostate cancer screening may be considered at age 40 years (see the NCCN Guidelines for Prostate Cancer Early Detection; available at www.nccn.org).

BARD1

A modest association between breast cancer and P/LP variants in the *BRCA1*-associated RING domain 1 (*BARD1*) gene has been found in case-control studies with a prevalence rate of 0.1% to 0.51% in patients with breast cancer. Studies show that *BARD1* is prevalent in 0.41% to 0.90% of patients with triple-negative breast cancer. The Breast Cancer Association Consortium and the CARRIERS case-control studies also found associations between a *BARD1* P/LP variant and increased risk of triple-negative breast cancer (0.42%; OR, 9.29; 95% CI, 4.58–18.85 and 0.41%; OR, 3.18; 95% CI, 1.16–7.42, respectively).

The panel recommends annual mammogram for carriers of a P/LP *BARD1* variant beginning at 40 years of age, with consideration of annual breast MRI. Age at which to initiate MRI in carriers with a P/LP *BARD1* variant depends on a number of risk factors, including family history, age, breast density, and patient preference. RRM is not recommended in carriers of a *BARD1* P/LP variant, but this procedure may be considered based on family history.

BRIP1

Panel testing of germline DNA in patients with ovarian cancer has shown that the prevalence rate of P/LP variants in the *BRCA1* interaction protein C-terminal helicase 1 gene (*BRIP1*), a Fanconi anemia gene, is about 1%. An analysis of 3236 females with epithelial ovarian cancer, 3431 controls, and 2000 unaffected high-risk females from an ovarian cancer screening trial (UKFOCSS) showed that *BRIP1* is associated with an increased risk for ovarian cancer ($P < .001$), with the RR for invasive epithelial ovarian cancer being 11.22 (95% CI, 3.22–34.10; $P < .001$) and 14.09 for high-grade serous disease (95% CI, 4.04–45.02; $P < .001$).³⁸³ A German study including 706 patients with ovarian cancer who do not carry a P/LP *BRCA1/2* variant showed a significant association between a *BRIP1* P/LP variant and ovarian cancer risk (OR, 20.97; 95% CI, 12.02–36.57; $P < .0001$), including late-onset ovarian cancer risk, with the highest OR for ovarian cancer diagnosed after age 60 (OR, 29.91; 95% CI, 14.99–59.66; $P < .0001$). An analysis of an Icelandic population (656 ovarian cancer cases, 3913 controls) also showed an association between *BRIP1* and increased risk for ovarian cancer (OR, 8.13; 95% CI, 4.74–13.95; $P < .001$). The cumulative lifetime risk of developing ovarian cancer by 80 years of age in carriers of a *BRIP1* P/LP variant is estimated to be 5.8% (95% CI, 3.6–9.1), though lifetime risk of developing ovarian cancer may also be as high as 12%.³⁵⁶ The panel recommends RRSO in carriers of a *BRIP1* P/LP variant at 45 to 50 years of age. A discussion about risk-reducing surgery may be

initiated earlier if there is a family history of early-onset ovarian cancer. Ultimately, large prospective trials are needed to make a firm age recommendation regarding when a discussion about RRSO should begin in these variant carriers.

Regarding breast cancer, a case-control study including 10,901 patients with triple-negative breast cancer showed that *BRIP1* was prevalent in 0.43% of cases. The panel has determined that more evidence is needed to provide breast screening recommendations in these carriers. *BRIP1* is associated with Fanconi anemia (group FANCF), inherited in an autosomal recessive manner. Therefore, counseling for carriers of *BRIP1* P/LP variants should include a discussion of reproductive options.

CDH1

Germline P/LP variants in *CDH1* are associated with hereditary diffuse gastric cancer and lobular breast cancer, and studies have reported a cumulative lifetime risk for breast cancer of 39% to 52%. Given the considerable risk for lobular breast cancer in carriers of a *CDH1* P/LP variant, the panel recommends screening with annual mammogram (or consideration of breast MRI) beginning at 30 years of age. Alternatively, screening may begin 5 to 10 years earlier than the youngest breast cancer diagnosis in the family. RRM may be discussed with these carriers.

There is controversy over how best to manage gastric cancer risk in individuals harboring a *CDH1* P/LP variant in the absence of a family history of gastric cancer. A small study found that more than half of the individuals with a *CDH1* P/LP variant who lacked a family history of gastric cancer had early-stage signet ring cell adenocarcinoma identified at the time of risk-reducing gastrectomy. A retrospective review including 75 families with a known *CDH1* P/LP variant showed that penetrance for lifetime risk of gastric cancer is associated with positive family history. See the NCCN Guidelines for Gastric Cancer (available at www.NCCN.org) for screening recommendations for gastric cancer for individuals with a *CDH1* P/LP variant. A report of two cases showed that *CDH1* P/LP variant may also be associated with cleft lip with or without cleft palate.

CHEK2

Another breast cancer susceptibility gene that has been identified is *CHEK2* (cell cycle checkpoint kinase 2). Studies investigating the association between primary breast cancer risk and specific *CHEK2* variants have primarily been based on the truncating variant 1100delC. An analysis from the Copenhagen General Population Study (N = 86,975) showed that *CHEK2* 1100delC heterozygotes had an increased risk for primary breast cancer when analyses were stratified by age and sex (HR, 2.08; 95% CI, 1.51–2.85). A

case-control study (10,860 cases and 9,065 controls) carried out by the CHEK2 Breast Cancer Case-Control Consortium of Europe and Australia showed that the 1100delC variant is associated with increased risk for primary breast cancer, even in females unselected for family history (OR, 2.34; 95% CI, 1.72–3.20; $P < .001$). Another case-control study (44,777 cases and 42,997 controls) showed that heterozygous 1100delC carriers have a significantly increased risk of developing ER-positive breast cancer (OR, 2.55; 95% CI, 2.10–3.10; $P < .001$), but not ER-negative breast cancer (OR, 1.32; 95% CI, 0.93–1.88; $P = 0.12$). Results from a meta-analysis including 18 case-control studies (26,336 cases and 44,219 controls) showed that the missense variant I157T is associated with a modestly increased risk for primary breast cancer (OR, 1.58; 95% CI, 1.42–1.75; $P < .001$). A retrospective cohort study including 3783 carriers of a *CHEK2* P/LP variant showed that primary breast cancer risk was elevated in carriers of c.444 + 1G>A (OR, 2.63; 95% CI, 1.59–4.35; $P < .001$), ex8_9del (OR, 2.36; 95% CI, 1.53–3.64; $P < .001$), p.R117G (OR, 1.65; 95% CI, 1.12–2.44; $P = .01$), and 1100delC variants (OR, 1.76; 95% CI, 1.55–2.00; $P < .001$), compared to *CHEK2* wild-type.

Ten-year cumulative risk of developing contralateral breast cancer in carriers of a *CHEK2* P/LP variant is 6–8%. The 15-year cumulative risk of developing contralateral breast cancer in premenopausal individuals who carry a *CHEK2* P/LP variant is 20.5% (95% CI, 8.9%– 47.4%), though this estimate is based on 7 cases and a wide confidence interval. The risk of metachronous contralateral breast cancer in women >65 years of age is not significantly different compared to non-carriers. Risk of contralateral breast cancer is higher if the primary breast cancer was ER-positive.

The 2022 comparative modeling analysis by Lowry et al showed that beginning annual MRI screening at age 30 to 35 years may reduce breast cancer mortality by more than 55% in carriers with a P/LP *CHEK2* variant. The panel also recommends annual mammogram for carriers of a P/LP *CHEK2* variant beginning at 40 years of age. Age at which to initiate MRI in carriers with a P/LP *CHEK2* variant depends on a number of risk factors, including family history, age, breast density, and patient preference. There are no data on the benefit of RRM for carriers of a P/LP *CHEK2* variant, but this procedure may be considered based on family history.

There is emerging evidence that *CHEK2* P/LP variants are associated with increased risk for prostate cancer. Prostate cancer screening may be considered at age 40 years (see the NCCN Guidelines for Prostate Cancer Early Detection; available at www.nccn.org).

MLH1, MSH2, MSH6, PMS2, EPCAM

Lynch syndrome results from a germline P/LP variant in 1 of 4 DNA mismatch repair (MMR) genes

(*MLH1*, *MSH2*, *MSH6*, or *PMS2*). Additionally, deletions in the *EPCAM* gene, which lead to hypermethylation of the *MSH2* promoter and subsequent *MSH2* silencing, cause Lynch syndrome. Females with Lynch syndrome are at heightened risk for endometrial cancer. With a lifetime risk of up to 60%, endometrial cancer is the second most common cancer in females with Lynch syndrome. For ovarian cancer, estimates vary depending on the specific gene, with risk estimates ranging from 4% to 20% for *MLH1*, 8% to 38% for *MSH2/EPCAM*, and $\leq 1\%$ to 13% for *MSH6*. Risk for ovarian cancer is not increased in carriers of a P/LP *PMS2* variant.

TVUS and serum CA-125 testing to screen for ovarian cancer in postmenopausal individuals has not been shown to be sufficiently sensitive or specific to warrant a routine recommendation. Since there is no effective screening for ovarian cancer, individuals should be educated on the symptoms that may be associated with the development of ovarian cancer, such as pelvic or abdominal pain, bloating, increased abdominal girth, difficulty eating, early satiety, or increased urinary frequency or urgency. Symptoms that persist for several weeks and are a change from baseline should prompt physician evaluation. Bilateral salpingo-oophorectomy (BSO) may reduce the incidence of ovarian cancer. The decision and timing of BSO as an option should be individualized based on whether childbearing is complete, menopausal status, comorbidities, family history, patient preference, and Lynch syndrome gene, as risks for ovarian cancer vary by mutated gene. Estrogen replacement after premenopausal oophorectomy may be considered. There is insufficient evidence to recommend RRSO in *MSH6* and *PMS2* P/LP variant carriers. Risk reduction agents should be considered, with detailed discussion between the physician and patient outlining the associated risks and benefits.

While studies have found that 42% to 51% of breast cancers in patients with Lynch syndrome are mismatch repair deficient (dMMR) with abnormal immunohistochemistry (IHC) corresponding to their germline pathogenic MMR gene variant, there are insufficient data supporting an increased risk for breast cancer for patients with Lynch syndrome.

Patients of reproductive age should be advised regarding their options for prenatal diagnosis and assisted reproduction, including PGT. This discussion should include known risks, limitations, and benefits of these technologies. If both partners are a carrier of a P/LP variant in the same MMR or *EPCAM* gene, then they should also be advised about the risk for constitutional MMR deficiency (CMMRD) syndrome, a rare recessive syndrome. More information regarding Lynch syndrome can be found in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal (available at www.NCCN.org).

NFI

Neurofibromatosis type 1 (NF1) is an autosomal dominant hereditary cancer syndrome that is caused by an *NF1* P/LP variant. NF1 is a neurocutaneous syndrome characterized by café-au-lait spots and axillary/inguinal freckling, associated with non-cancerous tumors of the nerve tissues. Individuals with NF1 have an increased risk for malignant peripheral nerve sheath tumors, other central nervous system (CNS) tumors, and gastrointestinal stromal tumors. A population-based study in Finland of 1404 patients with NF1 showed an estimated lifetime cancer risk of 59.6%. This study showed a significant association between NF1 and increased risk for breast cancer (SIR, 3.04; 95% CI, 2.06–4.31; $P < .001$). Among patients with breast cancer, NF1 was associated with poorer survival, with 5-year survival rates for patients with NF1 being 67.9%, compared to 87.8% in patients without NF1. Excess incidence was highest in females <40 years of age (SIR, 11.10; 95% CI, 5.56–19.50; $P < .001$). A population-based study in England of 848 patients with NF1 also showed an increased risk for breast cancer (SIR, 3.5; 95% CI, 1.9–5.9), especially among females <50 years (SIR, 4.9; 95% CI, 2.4–8.8).

Given the increased risk for early-onset breast cancer in carriers of these P/LP variants, annual breast screening with mammography should begin at 30 years of age. Screening with breast MRI could also be considered. The presence of neurofibromas in the breast may lead to false-positive MRI results, but more data are needed to determine the sensitivity and specificity of breast MRI in individuals with NF1. A prospective study of patients with NF1 from the United Kingdom (N = 448) showed that breast cancer risk in carriers of these P/LP variants is not significantly increased at ≥ 50 years of age. Case-control analyses of females with NF1 from England showed that RR estimates for women aged 30 to 39 years was 6.5 (95% CI, 2.6–13.5) and 4.4 for women aged 40 to 49 years (95% CI, 2.5–7.0). RR estimates then drop for women aged 50 to 59 years (RR, 2.6; 95% CI, 1.5–4.2) and continue to drop as age increases (RR, 1.9; 95% CI, 1.0–3.3 for women aged 60–69 years and RR, 0.8; 95% CI, 0.2–2.2 for women aged 70–79 years). These studies show that, beginning at age 50, breast cancer risk in women with NF1 may not significantly differ from that of women in the general population. Therefore, breast MRI screening in patients with NF1 may be discontinued at 50 years of age. There are no data regarding the benefit of RRM for carriers of *NF1* P/LP variants. Therefore, RRM is not recommended in these patients, but this procedure may be considered based on family history. Complications related to NF1 (e.g., neurologic complications) may appear early in life, and these have the potential to be severe. Therefore, referral to a neurofibromatosis specialist for management is recommended.

PALB2

PALB2 (partner and localizer of *BRCA2*) is a Fanconi anemia gene. *PALB2* P/LP variants are associated with increased risk for breast cancer, with studies of patients with breast cancer showing that 0.4% to 3% harbor a *PALB2* P/LP variant. A meta-analysis of three studies estimated an RR of 5.3 (90% CI, 3.0–9.4), while the most robust analysis to date included 524 families with a known P/LP *PALB2* variant and estimated an RR of 7.18 (95% CI, 5.82–8.85) for female breast cancer. The Breast Cancer Association Consortium study, CARRIERS study, and BRIDGES study all showed associations between a *PALB2* P/LP variant and increased risk of triple-negative breast cancer. *PALB2* P/LP variant is associated with a 41% to 60% lifetime risk of breast cancer. However, an analysis of 251 females who tested positive for ≥ 1 P/LP variant in a breast cancer susceptibility gene showed that the cumulative lifetime risk for breast cancer in individuals with an *PALB2* P/LP variant was lower, at 29.4% by age 70 years. The risk increases with increasing number of relatives showed that lifetime risk of breast cancer is as high as 76% when there is a family history of two first-degree relatives with breast cancer.

In a study of patients with breast cancer from Poland who underwent genetic testing, contralateral breast cancer was reported in 10% of *PALB2* carriers. This study also showed that 10-year survival among *PALB2* carriers with breast cancer was 48%, compared to 72% in carriers of a *BRCA1* P/LP variant and 75% in non-carriers ($P < .001$). The cumulative lifetime risk for male carriers of a P/LP *PALB2* variant is 0.9% (by age 70).

Ten-year cumulative risk of developing contralateral breast cancer in carriers of a *PALB2* P/LP variant is 5% to 8%. The 15-year cumulative risk of developing contralateral ER-negative breast cancer in premenopausal individuals who carry a *PALB2* P/LP variant is 35.5% (95% CI, 15.0%–84.0%), though this estimate is based on 4 cases and a wide confidence interval. The risk of contralateral breast cancer in carriers of a *PALB2* P/LP variant is only elevated in ER-negative disease. The risk of metachronous contralateral breast cancer in women >65 years of age is also not significantly different compared to non-carriers.

The panel recommends annual mammogram for carriers of a *PALB2* P/LP variant assigned female at birth beginning at 30 years of age. Breast MRI screening may also be considered. RRM for carriers of a *PALB2* P/LP variant may be considered. For individuals assigned male at birth, breast cancer screening similar to that for carriers of a *BRCA1* P/LP variant is reasonable.

PALB2 is associated with Fanconi anemia, inherited in an autosomal recessive manner. Therefore, counseling for carriers of *PALB2* P/LP variants should include a discussion of reproductive options.

RAD51C and RAD51D

Genes in the *RAD51* protein family are involved in homologous recombination and DNA repair. *RAD51C* and *RAD51D* have been shown to be associated with increased risk for ovarian cancer. Panel testing of germline DNA in females with ovarian cancer has shown that the prevalence rate of the *RAD51C* or *RAD51D* P/LP variant is about 1%. In a comparison of 1132 probands with a family history of ovarian cancer and 1156 controls, *RAD51C* was associated with an increased risk for ovarian cancer (RR, 5.88; 95% CI, 2.91–11.88; $P < .001$). Analyses from the same trial (911 probands and 1060 controls) also showed an association between *RAD51D* and increased risk for ovarian cancer (RR, 6.30; 95% CI, 2.86–13.85; $P < .011$). In a case-control analysis of 3429 females with epithelial ovarian cancer and 2772 controls, both *RAD51C* (OR, 5.2; 95% CI, 1.1–24; $P = .035$) and *RAD51D* (OR, 12.0; 95% CI, 1.5–90; $P = .019$) were associated with an increased risk for ovarian cancer. A study including 6178 and 6690 families with a known P/LP *RAD51C* and *RAD51D* variant, respectively, showed that the cumulative risk of developing ovarian cancer by age 80 was 11% for carriers of a *RAD51C* P/LP variant and 13% for carriers of a *RAD51D* P/LP variant.

The panel recommends RRSO in carriers of *RAD51C* and *RAD51D* P/LP variants starting at 45 to 50 years of age. A discussion about risk-reducing surgery may be initiated earlier if there is a family history of early-onset ovarian cancer. As with *BRIP1* P/LP variants, large prospective trials are needed to make a firm age recommendation regarding when a discussion about RRSO should begin in carriers of *RAD51C* and *RAD51D* P/LP variants is reasonable.

Some studies suggest an association between *PALB2* and increased ovarian cancer risk. The most

robust data to date showing an association between *PALB2* and increased ovarian cancer risk come from the international study, which included 524 families with a known P/LP *PALB2* variant. This study showed a 5% lifetime risk of ovarian cancer in carriers of a *PALB2* P/LP variant. RRSO may be considered in carriers of a *PALB2* P/LP variant starting at 45 to 50 years of age.

Regarding breast cancer, studies have shown prevalence rates of 0.23% to 0.45% for *RAD51C* and 0.29% to 0.38% for *RAD51D* in patients with triple-negative breast cancer. Two large case-control analyses showed that both *RAD51C* and *RAD51D* P/LP variants were significantly associated with triple-negative disease. The Breast Cancer Association Consortium study and the CARRIERS study showed associations between increased risk for ER-negative breast cancer and both *RAD51C* P/LP variant (OR, 3.99; 95% CI, 2.20–7.26 and OR, 2.19; 95% CI, 0.97–4.49, respectively) and *RAD51D* P/LP variant (OR, 2.92; 95% CI, 1.47–5.78 and OR, 3.93; 95% CI, 1.40–10.29, respectively), with prevalence rates of 0.26% and 0.24% for *RAD51C*, respectively, and 0.17% and 0.18% for *RAD51D*, respectively. The panel recommends annual mammogram for carriers of a P/LP *RAD51C* and *RAD51D* variant beginning at 40 years of age, with consideration of annual breast MRI. *RAD51C* is associated with Fanconi anemia, inherited in an autosomal recessive manner. Therefore, counseling for carriers of a *RAD51C* P/LP variant should include a discussion of reproductive options.

STK11

Germline *STK11* P/LP variants are associated with Peutz-Jeghers syndrome (PJS), an autosomal dominant disorder characterized by gastrointestinal polyps, mucocutaneous pigmentation, and elevated risk for gastrointestinal cancers as well as breast and non-epithelial ovarian cancers, such as Sertoli-Leydig tumors. Breast cancer risk in females with PJS is 8% at 40 years of age, 13% at 50 years of age, 31% at 60 years of age, and 45% at 70 years of age. Though there are no data on the benefit of RRM for carriers of *STK11* P/LP variants, RRM may be considered for these patients. Absolute risk of developing non-epithelial ovarian cancer (sex cord with annular tubules) is 18% to 21%. Information regarding screening for patients with PJS can be found in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal (available at www.NCCN.org).

Emerging Evidence

A systematic review of 21 papers including 47 patients with biallelic P/LP variants in *NTHL1* showed that 55% of the female patients were diagnosed with breast cancer. Another study including 29 carriers of biallelic *NTHL1* P/LP variants showed that 60% of females were diagnosed with breast cancer. Though breast cancer risk may be elevated, the evidence currently does not support screening beyond that which is recommended for the general population.

In a study including 3422 patients with breast or ovarian cancer who underwent tumor and germline sequencing, a *RAD51B* P/LP variant was found in 0.26%, which is comparable to prevalence rates for *RAD51C* and *RAD51D*. Breast screening may be considered in these carriers.

NCCN Genetic Testing Criteria

The NCCN genetic testing criteria for high-penetrance breast, ovarian, pancreatic, and prostate cancer are organized into three sections: 1) testing is clinically indicated; 2) testing may be considered; and 3)

there is a low probability of testing results having documented clinical utility (i.e., finding of high-penetrance genes). The testing criteria listed are for cancer susceptibility genes with strong or moderate evidence of actionability for breast, ovarian, pancreatic, and prostate cancer (e.g., *BRCA1/2*, *CDH1*, *PALB2*, *PTEN*, *STK11*, and *TP53* for breast cancer; additionally, testing criteria for LFS and Cowden syndrome continue to be contained in their own dedicated sections; see below). Included genes may change with emerging clinical data. Further, the personal and/or family history criteria included may suggest the possibility of additional syndromes and would necessitate additional unlisted genes to be evaluated.

The NCCN Panel recommends that individuals from a family with a known P/LP variant in a breast, ovarian, pancreatic, and/or prostate cancer susceptibility gene be tested for the known variant. However, multigene panel testing is often indicated in these individuals if the family history suggests a different syndrome in addition to the known variant. In individuals from a family without a known P/LP variant, germline multigene testing is recommended for those individuals who meet the testing criteria described in the *Hereditary Cancer Testing Criteria* section in the algorithm. Multi-gene testing may be considered for individuals who meet testing criteria and who previously underwent single-gene and/or absent deletion duplication analysis but tested negative. Both first- and second-degree relatives of individuals who meet these testing criteria are also eligible for testing, except for second-degree relatives of individuals with pancreatic cancer or prostate cancer, for whom prior probability of a high-penetrance cancer susceptibility gene is low in the absence of additional family history of cancer; only first-degree relatives of these affected individuals should be offered testing, unless indicated based on additional family history.

Testing Criteria Related to Prostate Cancer

Approximately 11% of patients with prostate cancer and at least 1 additional primary cancer carry germline P/LP variants associated with increased cancer risk. As described above, germline *BRCA1/2* P/LP variants are associated with increased risk for prostate cancer (see *BRCA-Related Breast/Ovarian Cancer Syndrome*, above). There is emerging evidence that *ATM* and *CHEK2* P/LP variants are associated with increased risk for prostate cancer. *HOXB13* P/LP variants have also been found in 1.4% to 4.5% of patients with prostate cancer. Prostate tumors with intraductal or cribriform histology may have increased prevalence of somatic MMR gene alterations. In addition, limited data suggest that germline homologous DNA repair gene mutations may be more common in prostate tumors of ductal or intraductal origin. Studies examining the association between carrying a germline *BRCA2* P/LP variant and intraductal histology have been conflicting. By definition, intraductal carcinoma includes cribriform proliferation of malignant cells, as long as they remain confined to a preexisting gland that is surrounded by basal cells. These features are seen frequently with an adjacent invasive cribriform component and would be missed without the use of basal cell markers.

Testing criteria related to prostate cancer include diagnosis of metastatic prostate cancer, as well as diagnosis of prostate cancer in an individual with Ashkenazi Jewish ancestry or suspicious family history (i.e., breast cancer that is triple-negative or diagnosed at an early age or in a male blood relative, ovarian cancer, pancreatic cancer, metastatic or high- or very-high-risk prostate cancer). Any patient in the high- or very-high-risk stratification group as defined in the NCCN Guidelines for Prostate Cancer (available at www.NCCN.org) is also eligible for testing without any additional testing criteria. Consistent with

the NCCN Guidelines for Prostate Cancer (available at www.NCCN.org), genetic testing may be considered in individuals diagnosed with intermediate-risk prostate cancer with intraductal/cribriform histology.

Systemic Therapy Decision-Making

Some of the NCCN treatment guidelines for *BRCA*-related cancers (Breast, Ovarian, Pancreatic Adenocarcinoma, Prostate; available at www.NCCN.org) recommend treatment with PARP (poly ADP-ribose polymerase) inhibitors for patients with germline or somatic *BRCA1/2* P/LP variants, as PARP inhibitors have been demonstrated to be active in these patients. These agents include olaparib and talazoparib for HER2- negative metastatic and as adjuvant treatment for high-risk HER2-negative breast cancer (olaparib only); niraparib, olaparib, and rucaparib for chemotherapy-refractory ovarian cancer; Olaparib and rucaparib for metastatic castration-resistant prostate cancer that has progressed following previous treatment; and olaparib and rucaparib as maintenance therapy options for metastatic pancreatic cancer. Even though the focus of these Guidelines continues to be on management of breast, ovarian, and/or pancreatic cancer risk in individuals with associated hereditary syndromes, the Guidelines now identify intent to aid in systemic therapy and surgical decision-making as a scenario in which germline testing is clinically indicated. If a P/LP variant is detected through tumor profiling that has clinical implications if identified in the germline, then germline testing for this variant is indicated.

Founder Mutations

The rate of the three founder P/LP variants in those of Ashkenazi Jewish ancestry is 2.2% to 2.5%. Studies have shown that genetic testing based on clinical guidelines emphasizing family history of breast, ovarian, pancreatic, prostate, or other cancers missed about 38% to 56% of P/LP variant carriers in those of Ashkenazi ancestry. Therefore, there is some evidence to support population-based genetic testing for individuals with Ashkenazi Jewish ancestry. However, there are concerns about the demand on genetic counseling resources, the preparedness of health care professionals to provide cancer genetic counseling and management, and participants' fears and concerns about testing, including those regarding privacy, stigmatization, and the need for appropriate medical and or surgical treatment in patients and family members found to have a founder P/LP variant. Thus, universal testing for founder *BRCA1/2* P/LP variants in individuals of Ashkenazi Jewish ancestry, regardless of personal or family history, should be offered primarily in the setting of longitudinal research studies. If there is no access to longitudinal studies, then testing may be offered when pre- and post-test genetic counseling are available (see above). There remains a vital need for longitudinal data from research studies exploring various methods of providing population- based genetic testing of individuals with Ashkenazi Jewish ancestry in the United States.

In addition to the *BRCA1* and *BRCA2* pathogenic variants (PVs) in those of Ashkenazi ancestry, there are other ancestries in which founder mutations have been identified. In these circumstances, the decision to test will depend on the prevalence of the PV in the local population, family history, clinical features, and age of cancer diagnosis. Examples where ancestry may, along with personal and/or family history, contribute to decisions about genetic testing include the following associations: numerous *BRCA1* and *BRCA2* PV in those of Spanish, Mexican, Central and South American descent; *BRCA1* PV and Polish ancestry; *BRCA1* and *BRCA2* PV and Bahamian ancestry; *BRCA2* PV and Icelandic ancestry; *BRCA1* and *BRCA2* PV in those of French Canadian ancestry; and *BRCA1* and *BRCA2* PV and

Hungarian ancestry. While emerging data derived from populations of Asian, Middle Eastern, and African origin have documented recurring mutations in *BRCA1* and *BRCA2* genes, population allele frequency data are not yet available to inform testing individuals based solely on ancestry in the absence of personal and/or family history. Other founder mutations that are not *BRCA1* and *BRCA2* include the *TP53* PV c.1010G>A (p. Arg337His) PV, which has been observed in a subset of those of Brazilian ancestry, and *CDKN2A* founder c.225_243del (p. Ala76fs) in those of Dutch ancestry.

Breast Cancer Population Testing

In 2019, the American Society of Breast Surgeons published a consensus statement recommending genetic testing for all patients with breast cancer. This recommendation was based on studies showing that criteria in testing guidelines miss some patients with breast cancer who harbor a P/LP variant and that population-based multi-gene testing is more cost-effective than testing based on personal and family history criteria. However, only 4.4% of patients with a high-penetrance mutation (i.e., *BRCA1/2*, *PALB2*, *TP53*, *PTEN*) were missed in the Beitsch et al study. Analyses from studies of postmenopausal patients with breast cancer showed rates of 3.6% to 5.6% harboring a P/LP variant. Further studies have reported that about 7% of those aged ≤ 65 years harbor a P/LP variant that is highly or moderately penetrant for breast cancer. A follow-up analysis of one of these studies examined age 60 as a cut-off for universal testing of patients with breast cancer and found that 8.2% of these patients harbor a P/LP variant associated with breast cancer. In this analysis, about 2% of patients diagnosed with breast cancer at age ≤ 60 years who did not meet other testing criteria harbored a highly penetrant P/LP variant associated with breast cancer. This percentage increased to about 5% when expanding the genes to include *ATM*, *CHEK2*, and *NF1*. It is not likely that patients diagnosed with breast cancer >60 years who do not meet other testing criteria will harbor a highly penetrant P/LP variant associated with breast cancer.

Additional tailoring of testing criteria in patients with breast cancer could be done based on histopathology or the presence of multiple primary breast cancers. An analysis of females >65 years ($N = 26,707$) from population-based case-control studies showed that 3.42% of females with ER-negative breast cancer and 3.01% of women with triple-negative breast cancer harbored a P/LP variant in a high-penetrance breast cancer susceptibility gene (*BRCA1*, *BRCA2*, and *PALB2*). Multiple studies also show that individuals with multiple primary breast cancers may be more likely than individuals with a single breast cancer to harbor a P/LP variant associated with breast cancer (7.1%–13.2% vs. 4.2%–9.4%). For the 2023 Guidelines update, the panel expanded the testing criterion for multiple primary breast cancers (synchronous or metachronous) to apply to all patients with breast cancer regardless of age of initial breast cancer diagnosis, which formerly applied only to patients with breast cancer diagnosed at ages 46 to 50 years.

The panel continues to endorse a risk-stratified approach and does not endorse universal testing of all patients with breast cancer due to limitations of this approach, such as low specificity, shortages in trained genetics health professionals to provide appropriate pre- and post-test genetic counseling, and lack of evidence to support risk management for genes included in many multi-gene panels. Though all patients with breast cancer should be evaluated to determine the appropriateness of germline genetic testing, testing should ultimately be based on patient

characteristics, such as those specified in the *Testing Criteria for High- Penetrance Breast Cancer Susceptibility Genes* in the algorithm.⁴⁹²

Li-Fraumeni Syndrome LFS is a rare hereditary cancer syndrome that is frequently associated with germline TP53 P/LP variants. The classic form of LFS is highly penetrant and characterized by a wide spectrum of neoplasms occurring at a young age and throughout the lifespan. An observational cohort study including 480 carriers with a TP53 P/LP variant enrolled in National Cancer Institute's (NCI) longitudinal Li-Fraumeni syndrome study showed that LFS is associated with a greater incidence of cancer than the general population (SIR, 23.9; 95% CI, 21.9–26.0), with the highest comparative incidence from childhood to age 30 years.⁵¹² An analysis from the NCI LiFraumeni Syndrome Study (N = 286) showed a cumulative lifetime cancer incidence of nearly 100%.⁵¹³ Soft tissue sarcomas, osteosarcomas (except Ewing sarcoma), pre-menopausal breast cancer, adrenocortical tumors, and brain tumors are referred to as the “core” cancers of LFS since they account for the majority of cancers observed in individuals with germline TP53 P/LP variants. In one study of 91 carriers of a germline TP53 P/LP variant, at least one of these cancers was found in one or more members of all families with a germline TP53 P/LP variant. Certain cancers are strongly associated with LFS, for example, hypodiploid acute lymphoblastic leukemia, choroid plexus tumor, and anaplastic rhabdomyosarcoma. Beyond the core cancers, LFS has also been associated with other leukemias, colon cancer, gastric cancer, bronchoalveolar and other lung cancers, prostate cancer, melanoma, and other CNS tumors. However, it is important to mention that estimations of cancer risks associated with LFS are limited to at least some degree by selection bias since dramatically affected kindreds are more likely to be identified and become the subject of further study. In addition, the majority of data are from LFS cohorts composed of self- identified white individuals.

Given the broad tumor types seen in LFS families, a number of different sets of criteria have been used to help identify individuals who have a high likelihood of having LFS including Classic, Chompret, Eeles and Birch criteria. For the purposes of the NCCN Guidelines, two of these criteria, Classic and Chompret, are used to facilitate the identification of individuals who are candidates for testing for TP53 P/LP variants.

Classic LFS criteria include, based on a study by Li and Fraumeni involving 24 LFS kindreds, a member of a kindred with a known TP53 P/LP variant; a combination of an individual diagnosed at ≤45 years of age with a sarcoma and a first-degree relative diagnosed with cancer at ≤45 years of age; and an additional first- or second-degree relative in the same lineage with cancer diagnosed at <45 years of age or a sarcoma diagnosed at any age (see *Testing Criteria for Li-Fraumeni Syndrome* in the algorithm). Classic LFS criteria have been estimated to have a high positive predictive value (estimated at greater than 50% and in >70% in some studies) as well as a high specificity, although the sensitivity is relatively low (estimated at 40%). Because individuals with TP53 P/LP variants are likely to have cancers beyond the LFS core cancers, the classic criteria will miss a significant portion of families.

Other groups have broadened the classic LFS criteria to facilitate identification of individuals with LFS. For example, criteria for TP53 testing proposed by Chompret and colleagues recommend testing for patients with multiple primary tumors of at least two “core” tumor types (i.e., sarcoma, breast cancer, adrenocortical carcinoma, brain tumors) diagnosed at <36 years of age or patients with

adrenocortical carcinoma diagnosed at any age, regardless of family history (see *Testing Criteria for Li-Fraumeni Syndrome* in the algorithm). The Chompret criteria have an estimated positive predictive value of 20% to 35%, and, when incorporated as part of *TP53* testing criteria in conjunction with classic LFS criteria, have been shown to improve the sensitivity to 95% (i.e., the Chompret criteria added to classic LFS criteria detected 95% of patients with *TP53* P/LP variants). Although not part of the original published criteria set forth by Chompret et al, the panel recommends adopting the 2015 Revised Chompret Criteria, including testing individuals with choroid plexus carcinoma or rhabdomyosarcoma of embryonal anaplastic subtype diagnosed at any age and regardless of family history (for inclusion in criterion 3), based on reports of considerable incidence of *TP53* P/LP variants found in patients with these rare forms of cancer. The panel supports the broader age cut-offs proposed by Tinat et al, based on a study in a large number of families, which detected germline *TP53* P/LP variants in affected individuals with later tumor onsets. These age cut-offs are: 1) individual diagnosed with LFS spectrum cancer <46 years of age who also have at least one FDR or SDR diagnosed with a LFS spectrum cancer <56 years of age or with multiple tumors; and 2) individual with multiple tumors from a LFS spectrum cancer, with the first diagnosed <46 years of age.

Patients with early-onset breast cancer (age of diagnosis ≤ 30 years) who were assigned female at birth, with or without family history of core tumor types, are another group for whom *TP53* gene P/LP variant testing may be considered. Several studies have investigated the likelihood of a germline *TP53* P/LP variant in this population. Among females <30 years of age with breast cancer and without a family history, the incidence of *TP53* P/LP variants has been reported at 3% to 8%. Some studies have also suggested that amplification of HER2 may arise in conjunction with germline *TP53* P/LP variants. *TP53* P/LP variants are a common finding across cancer types on tumor- only genomic testing but usually do not warrant consideration of germline testing. A recent analysis showed that the germline conversion rate (GCR, defined as the fraction of patients with a *TP53* P/LP variant on tumor-only testing that is determined to be germline on blood or saliva based genetic testing) of *TP53* P/LP variants was only 0.9% for tumors from all age patients and 5.1% for patients age <30. In the absence of paired germline analysis, germline testing should be offered if the personal or family history provides sufficient clinical suspicion of a germline P/LP variant. Consistent with European Society of Medical Oncology (ESMO) recommendations, the Panel also recommends that germline testing be offered to all patients who were diagnosed with any cancer at age <30 years.

Lastly, when *TP53* is included on multigene germline panels, the NCCN testing criteria for LFS are often not met. It has been argued that a spectrum of heritable *TP53*-related cancer syndromes exists. One study described families with *TP53* P/LP variants falling into a spectrum from classic LFS to attenuated families who do not meet criteria. These definitions will help future LFS research describe the populations being studied, but at this time LFS management is recommended for all individuals with P/LP *TP53* variants regardless of the presentation in the family.

If a *TP53* P/LP variant is found in blood, saliva, or buccal samples, especially in individuals whose personal or family history does not meet LFS criteria, this warrants consideration of testing of an alternative tissue, usually cultured skin fibroblasts, and close relatives to try to distinguish between germline, constitutional mosaicism, and somatic findings, such as CH or tumor contamination of peripheral blood. For patients who have a personal history of cancer, the panel also recommends looking for signs of tumor somatic interference and technical limitations.

A table describing workup and management depending on etiology of *TP53* P/LP variant found on genetic testing was added to the Guidelines for the 2024 update. The intent of this new table was to expand on potentially mosaic *TP53* findings. Prior to publication of this table, the Guidelines did not sufficiently point out the possibility of post-zygotic (somatic or constitutional) mosaicism (PZM) versus abnormal clonal expansions (ACE; including CHIP and clonal cytopenia of undetermined significance [CCUS]) and did not provide adequate guidance regarding how to care for these patients. A review of 84 *TP53*-positive probands identified through multigene testing on blood or saliva from 2012 to 2019 showed constitutional mosaicism in 8.3%. Ancillary tissue testing and cascade testing of children in all PZM and ACE *TP53* P/LP variant carriers is recommended, as this will further facilitate diagnosis and management. In addition, the clinical features that suggest CH versus PZM when a *TP53* P/LP variant is in the range of 30% to 70% variant allele frequency (VAF), in a patient with no prior chemotherapy and no hematologic abnormality, continue to be unknown.

Risk Assessment, Counseling, and Management

Discussions with patients about LFS management should address the limitations of screening for the many cancers associated with this syndrome. It is also important to address the psychosocial and quality-of-life aspects of this syndrome. Given the complexity of LFS management and that LFS is rare, individuals with LFS should be followed at centers with expertise in management of this syndrome. Personal and family history of cancer should be taken into consideration for screening (i.e., specific screenings, 5 to 10 years before earliest diagnosis). It is also important for patients' primary care providers and/or pediatricians to be informed about patients' diagnoses of LFS. Patients should be advised about the risk to relatives, and genetic counseling for relatives is recommended. For the 2024 Guidelines update, the panel added a section on pediatric surveillance in LFS.

Breast screening in adults with LFS includes clinical examination, breast imaging (MRI and mammogram, as indicated), and breast awareness. Although there are no data regarding risk reduction surgery, individuals with LFS who were assigned female at birth have increased breast cancer risk that warrants consideration of RRM. Given the high risk of contralateral breast cancer in LFS, the option of contralateral RRM should be discussed with patients diagnosed with breast cancer. Counseling for risk-reducing surgeries may include discussion of extent of cancer risk reduction/protection, risks associated with surgeries, degree of age-specific cancer risk, reconstructive options, and competing risks from other cancers. Family history and life expectancy should also be considered.

Use of a screening protocol that includes MRI may improve early cancer detection in individuals with LFS. Whole-body MRI for screening of cancers associated with LFS continues to be evaluated in multiple international trials. Use of whole-body MRI is appealing due to its wide anatomic coverage, lack of radiation and the potential to reduce the number of imaging studies that a patient undergoes. A meta-analysis including 578 individuals with *TP53* P/LP variants across 13 prospective cohorts showed that baseline whole-body MRI identified cancer in 7% of the sample, with 83% of the cancers being localized and able to treat with curative intent. In a prospective observational study, a clinical surveillance protocol for carriers of a *TP53* P/LP variant from families affected by LFS was incorporated. Eleven-year follow-up of this study, which included 89 carriers of a *TP53* P/LP variant, showed that this surveillance protocol may be beneficial, with 84% (16 of 19) of patients who were

diagnosed with cancer while under surveillance being alive at final follow-up, compared to 49% (21 of 43) of patients who were not being surveilled and were diagnosed with cancer due on symptoms ($P = .012$). Five-year OS was greater for patients undergoing surveillance (88.8%) compared to patients not undergoing surveillance (59.6%; $P = .013$). Based on these study results, the panel recommends annual whole-body MRI. It is important to note that, to date, data on the effectiveness of whole-body MRI have come from centers performing a high volume of these cancer screenings. Also, whole-body MRI protocols may vary. The Panel acknowledges that this surveillance method may not be uniformly available or affordable. Patients who do not have access to whole-body MRI should be encouraged to enroll in clinical trials and to work with their clinicians to develop an alternative screening program based on available cancer screening approaches. The panel also acknowledges that whole-body MRI screening of all individuals with LFS may result in false positives and overdiagnosis. Further, the utility of whole-body MRI has not been evaluated in individuals with a *TP53* P/LP variant who do not have a classic family history of LFS, a group that is increasingly being identified through multi-gene testing. The brain may be examined as part of whole-body MRI or as a separate exam.

In addition to whole body MRI, the panel recommends additional screening modalities for certain cancers. Individuals assigned female at birth should begin breast cancer screening with annual clinical breast exam and breast MRI at age 20 and the addition of annual mammogram at age 30. The panel recommends colonoscopy and upper endoscopy every 2 to 5 years starting at 25 years or younger in the context of prior abdominal radiation or family history. Dermatological exams are recommended. Finally, prostate cancer screening with PSA is recommended beginning at age 40.

Many of the other cancers associated with germline *TP53* P/LP variants do not lend themselves to early detection. Thus, additional recommendations for adults with LFS are general and include comprehensive physical examinations (including neurologic examination) every 6 to 12 months, especially when there is a high index of suspicion for second malignancies in cancer survivors and rare cancers. Clinicians should address screening limitations for other cancers associated with LFS. Screening methods for other LFS-associated cancers include periodic colonoscopy and upper endoscopy, dermatologic examination, and PSA. Education regarding signs and symptoms of cancer is important. Cancer screening in LFS should take into account prior treatment with radiation therapy.

Individuals with a *TP53* P/LP variant are at increased risk of second malignant neoplasms. Radiosensitivity in individuals with a *TP53* P/LP variant is not significantly different than in the general population, but carriers seem to be more susceptible to radioresistance. Though use of therapeutic RT should generally be avoided in individuals with a *TP53* P/LP variant, clinical decision-making should take into account the availability of other curative treatment options.

There is little evidence regarding care of *TP53* P/LP variant carriers with PZM or hypomorphic variants. Until there are more data on these carriers, they should be cared for as LFS, as opposed to patients with *TP53* CH, which should not be managed as LFS. Instead, given *TP53* mutation is considered a high-risk clinical feature in CH, patients with *TP53* CH may be referred to hematology expertise.

Cowden Syndrome/PTEN Hamartoma Tumor Syndrome

The spectrum of disorders resulting from germline P/LP variants in *PTEN* are referred to as PHTS. The spectrum of PHTS includes Cowden syndrome, Bannayan-Riley-Ruvalcaba syndrome (BRRS), adult Lhermitte-Duclos disease (LDD), Proteus-like syndrome, and autism spectrum disorders with macrocephaly. Cowden syndrome is rare, with an incidence of 1 in 200,000, although it is likely to be underestimated due to difficulties associated with making a clinical diagnosis of the disease. Cowden syndrome is an autosomal dominant disorder, and most cases are associated with germline *PTEN* P/LP variants, though one study found that germline *KILLIN* methylation may also be associated with this syndrome. The frequency of germline *PTEN* P/LP variant in Cowden syndrome cases is high, at approximately 80%.

Hamartomas (benign tumors resulting from an overgrowth of normal tissue) are a common manifestation of the PHTS syndromes. Cowden syndrome is associated with multiple hamartomatous and/or cancerous lesions in various organs and tissues, including the skin, mucous membranes, breast, thyroid, endometrium, and brain. However, it has been suggested that patients with other PHTS diagnoses associated with *PTEN* P/LP variants should be assumed to have Cowden syndrome-associated cancer risks.

The lifetime risk for breast cancer for females diagnosed with Cowden syndrome/PHTS has been estimated at 40% to 60%, with an average age of 38 to 50 years at diagnosis. Some studies (as discussed above) have reported a higher cumulative lifetime risk for breast cancer (77%–85%) in individuals with Cowden syndrome/PHTS or *PTEN* P/LP variants. A large European cohort study estimated lifetime risk for breast cancer in females as 54.3% to 75.8%. There have been only two cases of breast cancer reported in males with Cowden syndrome/PHTS. Although many females with Cowden syndrome/PHTS experience benign breast disease, there is no evidence that the rate is higher than in the general population.

Thyroid disease, including benign multinodular goiter, adenomatous nodules, and follicular adenomas, has been reported to occur in approximately 30% to 68% of adults with *PTEN* P/LP variants, and the lifetime risk for thyroid cancer (follicular or papillary) has been estimated at 3% to 16.5%. However, data tend to be aggregated, so it is difficult to calculate rates for multinodular goiter versus solitary nodules. A retrospective chart review of 47 children with *PTEN* P/LP variants showed that 26% had abnormal thyroid imaging. The youngest reported case of thyroid cancer in a child with Cowden syndrome/PHTS was at age 7.

Macrocephaly (defined as head circumference greater than the 97th percentile) is a common finding in patients with Cowden syndrome/PHTS. It has been estimated that approximately 80% to 100% of individuals with this syndrome will exhibit this clinical finding. Adult LDD and autism spectrum disorder characterized by macrocephaly are strongly associated with Cowden syndrome/PHTS. A rare, slow-growing, benign hamartomatous lesion of the brain, LDD, is a dysplastic gangliocytoma of the cerebellum. In a multicenter prospective study examining 3042 probands who met clinical criteria for Cowden syndrome/PHTS, 6% met criteria for LDD. In a study of individuals meeting the diagnostic criteria for Cowden syndrome/PHTS, the cumulative lifetime risk for LDD was reported to be 32%. The preponderance of evidence supports a strong association between adult-onset LDD and the presence of a *PTEN* P/LP variant, although exceptions have been reported. In addition, there is a relatively large body of evidence to support that 10% to 20% of individuals with autism spectrum

disorder and macrocephaly carry germline *PTEN* P/LP variants.

As in many other hereditary cancer syndromes, affected individuals are more likely to develop bilateral and multifocal cancer in paired organs. Although not well defined, females with Cowden syndrome/PHTS may have a 5% to 22% risk for endometrial cancer. While many females with Cowden syndrome/PHTS may also have uterine fibroids, this risk is not likely to be much greater than in females without Cowden syndrome/PHTS or *PTEN* P/LP variant.

In addition, brain tumors and vascular malformations affecting any organ are occasionally seen in individuals with Cowden syndrome/PHTS, although the risks for developing these conditions are not well defined. It is important to note, however, that most of the data on the frequencies of the clinical features of Cowden syndrome/PHTS are from compilations of case reports of relatively young individuals who may have subsequently developed additional signs of the disease (i.e., new cancerous lesions), and these data are also likely to be confounded by selection bias. Furthermore, a considerable number of these studies were published prior to the establishment in 1996 of the International Cowden Consortium operational diagnostic criteria for the syndrome, which were based on published data and the expert opinion of individuals representing a group of centers mainly in North America and Europe.

Benign skin lesions are experienced by most to all patients with Cowden syndrome/PHTS. Skin lesions associated with Cowden syndrome/PHTS include trichilemmomas (i.e., benign tumors derived from the outer root sheath epithelium of a hair follicle), oral papillomas, mucocutaneous neuromas (hamartoma of the peripheral nerve sheath), palmoplantar keratoses, penile pigmentation, lipomas and vascular anomalies, and fibromas. Trichilemmomas associated with Cowden syndrome/PHTS tend to appear on the face, particularly the eyes, mouth, nose, and forehead. Most individuals with Cowden syndrome/PHTS exhibit characteristic mucocutaneous lesions by their twenties, and such lesions have been reported to occur in 99% of individuals with Cowden syndrome/PHTS, showing nearly complete penetrance, although this may be a reflection of selection bias in the cases reported. The presence of three or more mucocutaneous neuromas is considered a major diagnostic criterion of Cowden syndrome/PHTS, while the presence of two or more trichilemmomas has been reported to be pathognomonic for Cowden syndrome/PHTS. However, since most of the evidence regarding trichilemmomas is from the older literature, it is possible that the association with Cowden syndrome/PHTS is somewhat overestimated. There are reports of individuals with a solitary trichilemmoma who do not have Cowden syndrome/PHTS. Nevertheless, due to the strong association between these lesions and Cowden syndrome/PHTS and the difficulty in clinically distinguishing between a trichilemmoma and another mucocutaneous lesion, it is important that a diagnosis of trichilemmoma is histologically confirmed.

It was previously estimated that about half of individuals with Cowden syndrome/PHTS have gastrointestinal polyps. However, this was almost certainly an underestimate. In an analysis of 67 *PTEN* P/LP variant carriers undergoing colonoscopy, colorectal polyps were found in 92.5% of patients. About half of the patients undergoing colonoscopy had hyperplastic polyps, and about 25% had polyps that were hamartomatous, ganglioneuromatous, or adenomatous. Adenomatous or hyperplastic polyps were associated with development of colorectal cancer in this sample. Out of 39 carriers of a *PTEN* P/LP variant undergoing esophagogastroduodenoscopy (EGD), upper gastrointestinal polyps were found in 67% of patients. A systematic review of published case series

(N = 102) regarding gastrointestinal manifestations in Cowden syndrome/PHTS and component syndromes showed that 92.5% of these patients had polyps, with 64% having 50 or more. Histologies were described as: hyperplastic (44%), adenomatous (40%), hamartomatous (38%), ganglioneuroma (33%), and inflammatory (24.5%). Other studies have also reported ganglioneuromatous polyps (i.e., rare, benign peripheral nervous system tumors) in this population. A retrospective chart review of 47 children with *PTEN* P/LP variants showed that only 13% had gastrointestinal polyps, but 34% had other gastrointestinal symptoms such as abdominal pain, rectal bleeding, and/or constipation. Early-onset (<50 years of age) colorectal cancer has been reported in 13% of patients with *PTEN* P/LP variant-associated Cowden syndrome/PHTS, suggesting that routine colonoscopy may be warranted in this population. The lifetime risk for colorectal cancer has been estimated as 9% to 16%.

Several studies have projected lifetime estimates of cancer risk that are significantly higher than previously estimated. In a study of patients meeting diagnostic criteria for Cowden syndrome/PHTS (N = 211; identified from published literature and records from a single institution), the cumulative lifetime risk for any cancer was 89%. *PTEN* P/LP variants had been identified in 97 of 105 patients (92%) who underwent testing. The cumulative lifetime cancer risks for all patients (n = 210) were 81% for female breast cancer, 21% for thyroid cancer, 19% for endometrial cancer, 15% for renal cancer, and 16% for colorectal cancer. In a prospective study that evaluated genotype-phenotype associations between *PTEN* P/LP variants and cancer risks, deleterious germline P/LP variants in *PTEN* were identified in 368 patients. Calculation of age-adjusted SIRs using cancer incidence data from the SEER database showed elevated SIRs among individuals with *PTEN* P/LP variants for breast cancer (25), thyroid cancer (51), endometrial cancer (43), colorectal cancer (10), renal cancer (31), and melanoma (8.5). The estimated cumulative lifetime cancer risks were 85% for breast, 35% for thyroid, 28% for endometrial, 9% for colorectal, 34% for renal, and 6% for melanoma. In another study in individuals with PHTS found to have deleterious germline *PTEN* P/LP variants (N = 154; detailed information available in n = 146), age- and gender-adjusted SIRs were elevated for female breast cancer (39), endometrial cancer (49), female thyroid cancer (43), male thyroid cancer (199.5), female melanoma (28), and male melanoma (39). The cumulative lifetime risks in these individuals were 77% for female breast cancer and 38% for thyroid cancer. The cumulative lifetime risk for any cancer was 85% overall, and females with Cowden syndrome/PHTS were found to have a 2-fold greater cancer risk compared with males with Cowden syndrome/PHTS. It is important to note, however, that all three of these studies suffer from significant ascertainment biases, in that patients were usually selected for *PTEN* testing based on the presence of these malignancies, which would inflate the projected lifetime cancer estimates. An observational study of 180 patients with *PTEN* P/LP variants used Kaplan-Meier methods to estimate that female carriers (n = 99) have an 87% cumulative risk of developing any cancer and/or LDD by 60 years of age, while male carriers have a cumulative risk of 56%.

The BRRS variant of Cowden syndrome/PHTS has been characterized by the presence of multiple lipomas, gastrointestinal hamartomatous polyps, macrocephaly, hemangiomas, developmental delay, and pigmented macules on the glans penis, although formal diagnostic criteria have not been established for this syndrome. *PTEN* gene P/LP variants testing in individuals characterized with BRRS have been reported in approximately 60% of these patients. Further, in another study, 10% of patients with BRRS for whom a *PTEN* P/LP variant test was negative were shown to be carriers of large *PTEN* gene deletions.

Risk Assessment, Counseling, and Management

The assessment of individuals suspected of having Cowden syndrome/PHTS incorporates both a history of the benign and malignant conditions associated with the syndrome and a targeted physical examination, including the skin and oral mucosa, breast, and thyroid gland and head circumference (see *Testing Criteria for Cowden Syndrome/PHTS* in the algorithm). The NCCN Guidelines Panel has established a list of criteria to help indicate which individuals are candidates for testing for *PTEN* P/LP variants (see *Testing Criteria for Cowden Syndrome/PHTS* in the algorithm). These criteria are used to assess the need for further risk assessment and genetic testing. When *PTEN* is included on multi-gene panels, these testing criteria do not need to be met. Clinical diagnostic criteria have also been developed to help identify clinical features associated with Cowden syndrome/PHTS (see *Revised Clinical Diagnostic Criteria for PTEN Hamartoma Tumor Syndrome* in the algorithm and discussed below under *Clinical Diagnostic Criteria*). Patients who meet clinical diagnostic criteria for Cowden syndrome/PHTS as described in this section are candidates for testing for *PTEN* P/LP variants.

Testing Criteria

Testing criteria for Cowden syndrome/PHTS are grouped into three general categories. A patient is considered for testing for *PTEN* P/LP variants based on whether they meet certain criteria or combinations of criteria from these three categories. The first criteria category includes individuals meeting diagnostic criteria for Cowden syndrome: a personal history of BRRS, adult LDD, autism spectrum disorder with macrocephaly, or two or more biopsy-proven trichilemmomas. Any individual presenting with one or more of these diagnoses warrants *PTEN* testing. Previously, some of the criteria from this group have been referred to as “pathognomonic,” although it is unlikely that any of these conditions can stand alone as a definitive diagnostic criterion for Cowden syndrome/PHTS. Another criterion that can be considered to be sufficient to warrant testing for *PTEN* P/LP variants is a family history that includes the presence of a known *PTEN* P/LP variant.

The next category of criteria represents “major” features associated with Cowden syndrome/PHTS and are described in the Guidelines (see *Testing Criteria for Cowden Syndrome/PHTS* in the algorithm). With respect to decisions related to the presence of mucocutaneous lesions, the panel did not consider the available literature to be adequate to accurately specify the number or extent of these lesions required for the condition to be defined as a major criterion for Cowden syndrome/PHTS, and clinical judgment is needed when evaluating such lesions. An individual exhibiting two or more major criteria where one criterion is macrocephaly meets the testing threshold. An individual with three or more major criteria (without macrocephaly) is also considered to meet the threshold for testing. In addition, individuals exhibiting one major criterion with three or more minor criteria (see *Testing Criteria for Cowden Syndrome/PHTS* in the algorithm) also meet the testing threshold; if an individual exhibits two or more major criteria but does not have macrocephaly, then one of the major criteria may be included as one of the three minor criteria to meet the testing threshold.

The final category of criteria represents features with a “minor” association with Cowden syndrome/PHTS. These criteria are described in the Guidelines (see *Testing Criteria for Cowden Syndrome/PHTS* in the algorithm). An individual would need to exhibit four or more minor criteria or as discussed above, three or more minor criteria and one major criterion to meet testing.

Lastly, an individual who has a first-degree relative diagnosed with Cowden syndrome/PHTS or BRRS for whom testing has not been performed would also meet the threshold for *PTEN* testing if the individual meets at least one major criterion or two or more minor criteria. *PTEN* P/LP variants are commonly found in tumor tissue. If a *PTEN* variant is detected through tumor profiling and would be classified as P/LP if present in the germline, then germline testing for *PTEN* should be considered.

Clinical Diagnostic Criteria

The frequency of *PTEN* P/LP variant in individuals meeting International Cowden Consortium diagnostic criteria for Cowden syndrome has previously been estimated at about 80%. However, evaluation of data based on samples analyzed at a single academic pathology laboratory (N = 802 evaluable) reported a much lower frequency (34%) of *PTEN* P/LP variants among individuals meeting diagnostic criteria for Cowden syndrome. The authors concluded that the current Consortium diagnostic criteria are not as sensitive in identifying individuals with *PTEN* P/LP variants as previously estimated. Since *PTEN* P/LP variants are relatively rare, recommendations regarding Cowden syndrome diagnostic criteria may be based on studies with a small number of patients. Studies with larger samples have their flaws as well, as patients are selected for testing based on the number and magnitude of clinical features, which may lead to overestimation of the features of Cowden syndrome. A review was conducted examining each consortium diagnostic criterion, and revised criteria were proposed that are more stringent and take into account clinical features that are often seen in PHTS. The criteria were designed by focusing on clinical features associated with *PTEN* P/LP variants. The panel recommends using these criteria for clinical diagnosis of PHTS (see *Revised Clinical Diagnostic Criteria for PTEN Hamartoma Tumor Syndrome* in the algorithm).

Screening Recommendations

Cancer is the major health risk associated with Cowden syndrome/PHTS. Therefore, the NCCN Panel has outlined guidelines for prevention and early detection screening of commonly associated cancers with Cowden syndrome/PHTS. Current medical management recommendations for Cowden syndrome/PHTS include annual physical examinations, starting at 18 years of age (or 5 years before the youngest age of diagnosis of a component cancer in the family).

The recommendations for individuals with Cowden syndrome/PHTS who were assigned female at birth focus on primary and secondary prevention options for breast cancer since this is the most commonly associated cancer in individuals with Cowden syndrome/PHTS based on the available literature. Individuals assigned female at birth should begin regular monthly breast self-examinations at 18 years of age and have a semi-annual clinical breast examination beginning at 25 years of age or 5 to 10 years earlier than the earliest known breast cancer in the family (whichever comes first). Individuals assigned female at birth should also have an annual mammogram and breast MRI screening with and without contrast starting at 30 years of age, or 10 years earlier than the earliest known breast cancer in the family (whichever comes first). After 75 years of age, management should be considered on an individual basis. In patients treated for breast cancer who were assigned female at birth and who have not had bilateral mastectomy, mammography and breast MRI screening with contrast should continue as recommended based on age. A single-center retrospective study including 65 females diagnosed with PHTS showed that the yield of breast cancer screening (MRI and mammography) is comparable in this population,

compared to other carriers of high-penetrance breast cancer susceptibility genes (e.g., *BRCA1/2*).

Although there are no data regarding risk reduction surgery in individuals with Cowden syndrome who were assigned female at birth, the option of RRM and hysterectomy should be discussed. Oophorectomy is not indicated for Cowden syndrome since ovarian cancer risk is not elevated in these patients. Counseling for risk-reducing surgeries may include syndrome/PHTS. Current medical management recommendations for Cowden syndrome/PHTS include annual physical examinations, starting at 18 years of age (or 5 years before the youngest age of diagnosis of a component cancer in the family).

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Although there are no data regarding risk reduction surgery in individuals with Cowden syndrome who were assigned female at birth, the option of RRM and hysterectomy should be discussed. Oophorectomy is not indicated for Cowden syndrome since ovarian cancer risk is not elevated in these patients. Counseling for risk-reducing surgeries may include discussion of extent of cancer risk reduction/protection, risks associated with surgeries, and reproductive options. It is also important to address the psychosocial and quality-of-life aspects of undergoing risk-reducing surgical procedures.

Given that Cowden syndrome is rare, there are no data on screening for endometrial cancer in these patients, though consideration of screening can begin as early as age 35. The panel recommends patient education regarding the symptoms of endometrial cancer including the necessity of a prompt response to symptoms such as abnormal bleeding. Prompt reporting promotes early detection of endometrial cancer. The evaluation of these symptoms should include an endometrial biopsy. Though endometrial cancer screening does not have proven benefit in individuals with Cowden syndrome, endometrial biopsy is highly sensitive and specific as a diagnostic procedure. Therefore, screening through endometrial biopsy every 1 to 2 years may be considered.

Routine TVUS to screen for endometrial cancer in postmenopausal individuals has not been shown to be sufficiently sensitive or specific to warrant a positive recommendation but may be considered at the clinician's discretion. However, TVUS is not recommended as a screening tool in premenopausal individuals due to the wide range of endometrial strip thickness throughout the normal menstrual cycle.

Individuals with Cowden syndrome/PHTS have approximately at least a 3% to 10% lifetime risk of developing thyroid cancer, compared to about 1% in the general population. An annual thyroid ultrasound should be performed, starting at age 7. Children at risk for a *PTEN* P/LP variant (based on a parent's carrier status) whose parents wish to delay genetic testing may also undergo annual thyroid ultrasound, since this is a non-invasive procedure. Colonoscopy is recommended starting at 35 years of age, or earlier if symptomatic. If a close relative was diagnosed with colon cancer before 40 years of age, then colonoscopy screening should begin 5 to 10 years before the age of the earliest known diagnosis. Colonoscopy should be performed every 5 years or more frequently in cases where the patient is symptomatic, or polyps are found. To screen for renal cell carcinoma, renal ultrasound should be considered every 1 to 2 years beginning at 40 years of age. Annual dermatologic examination is recommended. If there are symptoms in children, then assessment of psychomotor abilities should be considered, as well as a brain MRI. Education regarding the signs and symptoms of cancer is important; patients should also be advised about the risk to relatives, and genetic counseling is recommended for at-risk relatives.

No published data exist on the use of prenatal diagnostics/genetic testing for *PTEN* P/LP variants in families with Cowden syndrome. For a general discussion on the topic of reproductive options and counseling considerations, see the Discussion section above on *Reproductive Options* under *Genetic Risk Assessment and Counseling*.

Hereditary Pancreatic Cancer

Pancreatic cancer is thought to have a familial or hereditary component in approximately 10% of cases. Harboring a P/LP variant has been found to be associated with a greater incidence of pancreatic cancer than family history alone (without the presence of an associated germline variant). An analysis of 250 patients with pancreatic cancer who underwent multigene panel testing with a >80 gene panel showed that 15% harbored a P/LP variant. Germline P/LP variants found in pancreatic adenocarcinoma include *BRCA1*, *BRCA2*, *CDKN2A*, MMR genes associated with Lynch syndrome (specifically *MSH2*, *MLH1*, *MSH6*, and *EPCAM*), *ATM*, *PALB2*, *STK11*, and *TP53*.

BRCA2 and *CDKN2A* are generally the most prevalent, with rates in moderate- to high-risk families ranging from 2% to 6% for *BRCA2* and 1.5% to 2.5% for *CDKN2A*. In addition, hereditary pancreatitis, which is associated with increased risk for pancreatic cancer, is associated with the genes *PRSS1* and *SPINK1*. Patients with pancreatic cancer and Ashkenazi Jewish ancestry may have a greater likelihood of testing positive for a *BRCA1/2* P/LP variant, with prevalence of detected P/LP variants in this group ranging from 5.5% to 19%, with P/LP variants being more common for *BRCA2*.

Given the considerable rate of predisposing P/LP variants in patients with pancreatic cancer, as well as the fact that typical clinical factors (e.g., young age of onset, family history of cancer) are poorly predictive for identifying carriers of a P/LP variant, universal genetic testing for these individuals is warranted. Given the elevated rates of P/LP variants in pancreatic cancer and that pancreatic cancer risk increases when there is a family history, testing of first-degree relatives of patients may be beneficial. However, testing the patient is preferred. Testing of second-degree relatives is generally not recommended but may be considered in select cases. Given that mortality rates for this cancer are high, it may be beneficial to family members to test patients near the time of diagnosis, since the option to

test the patient may not be available for very long. Family history of pancreatic cancer with unknown histology is often presumed to be exocrine. Detecting a germline P/LP variant can potentially aid in treatment decision-making, particularly regarding systemic therapy options (see *Systemic Therapy Decision-Making* above).

Pancreas Screening

Evidence to support screening for pancreatic cancer comes from studies including those who harbor an associated germline P/LP variant and/or those who have a particularly strong family history of pancreatic cancer (at least one first-degree relative and at least one second-degree relative on the same side of the family). The multicenter CAPS5 prospective cohort study, which included 1461 individuals considered high-risk (i.e., P/LP carriers of *CDKN2A*, *STK11*, *ATM*, *BRCA1*, *BRCA2*, *MSH2*, *MLH1*, *MSH6*, *EPCAM*, or *PALB2*; or family history of ≥ 1 first-degree and 1 second-degree relative with pancreatic cancer), evaluated stage at diagnosis and outcome of individuals diagnosed with pancreatic cancer who underwent an annual pancreas imaging surveillance protocol. Out of 10 patients diagnosed with pancreatic cancer, seven were diagnosed with stage I disease. Median OS was significantly greater in patients diagnosed with screening-detected pancreatic cancer, compared to patients diagnosed outside of the surveillance protocol (9.8 years vs. 1.5 years, respectively; HR, 0.13; 95% CI, 0.03–0.50; $P = .003$). An analysis of outcomes from three European centers including 411 asymptomatic individuals showed that pancreatic cancer was detected in 7% of carriers of a *CDKN2A* P/LP variant and less than 1% of those with familial pancreatic cancer.⁴⁸⁷ For the carriers of a *CDKN2A* P/LP variant for whom a lesion was detected, 75% were resectable, with a 5-year OS rate of 24%. A prospective study including 347 carriers of a germline *CDKN2A* P/LP variant who participated in a 20-year pancreatic cancer surveillance protocol at a medical center in the Netherlands showed that pancreatic adenocarcinoma was diagnosed in 20.7% by age 70 years. Out of the 36 pancreatic cancers diagnosed, 83.3% were resectable, and 33.3% were diagnosed as stage I. Five-year OS in those who underwent resection was 44.1% (95% CI, 27.2–71.3). In another analysis from six high-volume centers in Italy including 187 high-risk individuals, abnormalities were detected in about 28%. Out of the cysts detected, 62.2% were branch-duct intraductal papillary mucinous neoplasms. Pancreatic adenocarcinomas made up 2.6% of the findings ($n = 5$). Finally, another analysis including screening of 354 asymptomatic high-risk individuals showed suspicious pancreas lesions in 19%. Out of the lesions detected from screening, 90% were resectable, and the 3-year OS rate was 85% in those with resectable lesions.

The considerable rate of resectable asymptomatic lesions found from routine screening of high-risk individuals demonstrates the potential for downstaging (i.e., identification of lesions at an earlier stage). There is also the potential for impact on mortality rates, though long-term studies are needed in this area. Lesions detected through routine screening may not always require resection (e.g., sporadic branch-duct intraductal papillary mucinous neoplasms). Although there is much more experience with evaluating and managing pancreatic cysts and other pancreatic imaging abnormalities, determination of the overall risk/benefits of pancreatic surveillance requires further study. Results of surveillance of high-risk individuals performed in tertiary care/high-volume centers under clinical trial settings may not be the same as those performed in routine clinical practice. Data are beginning to better define which screen-detected lesions in high-risk individuals should be considered to be at particularly high risk for neoplastic progression (e.g., those with a solid pancreatic

mass, those with pancreatic duct abnormalities, those with growing pancreatic cysts), but further data are needed to better define the threshold for surgical intervention in high-risk individuals undergoing pancreatic cancer screening.

With the exception of *CDKN2A* and *STK11*, pancreas cancer screening in individuals who have a P/LP variant associated with increased risk of exocrine pancreatic cancer (i.e., *ATM*, *BRCA1*, *BRCA2*, *MSH2*, *MLH1*, *MSH6*, *EPCAM*, *PALB2*, *TP53*) is not recommended unless there is additional family history of pancreatic cancer (at least 1 first- or second- degree relative). If family history criteria are met, then pancreas screening may be considered at age 50, or 10 years younger than the earliest pancreatic cancer diagnosis in the family, whichever is earlier. The International Cancer of the Pancreas Screening Consortium recommendations for pancreas screening in individuals with increased risk for hereditary pancreatic cancer do not include carriers of a *TP53* P/LP variant in this group, as there are very limited data on pancreatic cancer screening in these carriers. However, the NCCN Guidelines Panel recommends that pancreatic cancer screening be considered in carriers of a *TP53* P/LP variant, if there is additional family history of pancreatic cancer (at least 1 first- or second-degree relative), as there is some evidence of a modestly increased risk of pancreatic cancer in these carriers.

For carriers of a *CDKN2A* or *STK11* P/LP variant, no additional family history is needed to warrant screening. For carriers of a *CDKN2A* P/LP variant, screening may be considered at age 40, or 10 years younger than the earliest pancreatic cancer diagnosis in the family, whichever is earlier. For carriers of a *STK11* P/LP variant, screening may be considered beginning at ages 30 to 35, or 10 years younger than the earliest pancreatic cancer diagnosis in the family, whichever is earlier.

Hereditary pancreatitis is defined by the presence of a causative P/LP variant such as *PRSS1* or *SPINK1*, or a suspicious family history of chronic pancreatitis (two first-degree relatives or three second-degree relatives across two or more generations) without precipitating factors and with a negative workup for other known causes of pancreatitis. Hereditary pancreatitis is associated with increased lifetime risk of exocrine pancreatic cancer. The clinical significance of the P/LP variant such as *PRSS1* or *SPINK1* is unclear without a clinical history of pancreatitis. Therefore, germline testing for *PRSS1*, *SPINK1*, and other genes associated with pancreatitis is generally not recommended unless one's personal or family history is suggestive of hereditary pancreatitis. Pancreas cancer screening is recommended in individuals harboring one of these variants only in the presence of a clinical phenotype consistent with hereditary pancreatitis. For individuals meeting these criteria, screening may begin at age 40, or 20 years after onset of pancreatitis, whichever is earlier.

When screening is recommended, it may be done with contrast-enhanced MRI/magnetic resonance cholangiopancreatography (MRCP) and/or endoscopic ultrasound (EUS). MRI and EUS have been shown to be superior in detection of subcentimeter pancreatic cysts, compared to CT. Screening at a high-volume center of expertise is recommended, preferably in the context of a research study. In those for whom screening shows potentially concerning features that suggest progression, shorter screening intervals may be indicated.

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